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Uni-Perceiver-MoE: Learning Sparse Generalist Models with Conditional MoEs

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Abstract

To build an artificial neural network like the biological intelligence system, recent works have unified numerous tasks into a generalist model, which can process various tasks with shared parameters and do not have any task-specific modules. While generalist models achieve promising results on various benchmarks, they have performance degradation on some tasks compared with task-specialized models. In this work, we find that interference among different tasks and modalities is the main factor to this phenomenon. To mitigate such interference, we introduce the Conditional Mixture-of-Experts (Conditional MoEs) to generalist models. Routing strategies under different levels of conditions are proposed to take both the training/inference cost and generalization ability into account. By incorporating the proposed Conditional MoEs, the recently proposed generalist model Uni-Perceiver can effectively mitigate the interference across tasks and modalities, and achieves state-of-the-art results on a series of downstream tasks via prompt tuning on 1% of downstream data. Moreover, the introduction of Conditional MoEs still holds the generalization ability of generalist models to conduct zero-shot inference on new tasks, *e.g.,* video-text retrieval and video caption. Code and pre-trained generalist models shall be released at https://github.com/fundamentalvision/Uni-Perceiver.

1 Introduction

Generalist models that handle multiple modalities and numerous tasks have been long pursued by the machine learning community. However, previous researches [65, 89, 71] focus on developing specialized models with task-specific modules. When these models are applied to new tasks, the specifically-designed components need to be redesigned on demand and fine-tuned on sufficient downstream data. As a result, their model size increases with the number of diverse downstream tasks, conflicting with the goal of generalist models.

Recently, some pioneers [93, 79, 3, 84, 86, 62] have made preliminary attempts to build generalist models by modeling various tasks into a unified formulation. With the unified modeling, large-scale pre-training on various datasets enables the generalist models to process different downstream tasks using shared parameters. These generalist models not only achieve competitive performance on pre-training tasks [79, 3, 84, 86], but also can perform zero-shot inference on novel tasks without introducing additional parameters [93, 62].

However, compared to specialized models with specific parameters for each task, generalist models with shared parameters would suffer from the task-interference issue — different tasks with shared

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Uni-Perceiver-MoE：使用条件 MoE 学习稀疏通用模型

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摘要

为了构建类似生物智能系统的人工神经网络，近期工作已将众多任务统一到一个通用模型中，该模型可以使用共享参数处理各种任务，并且不包含任何任务特定模块。尽管通用模型在各种基准上取得了有前景的结果，但与任务专用模型相比，它们在某些任务上的性能有所下降。在这项工作中，我们发现不同任务和模态之间的干扰是导致这一现象的主要因素。为缓解这种干扰，我们将条件混合专家（Conditional Mixture-of-Experts，Conditional MoEs）引入通用模型。我们提出了不同条件层级下的路由策略，以同时兼顾训练/推理成本和泛化能力。通过结合所提出的 Conditional MoEs，近期提出的通用模型 Uni-Perceiver 能够有效缓解跨任务和跨模态的干扰，并通过在 1% 下游数据上进行提示调优，在一系列下游任务上取得最先进的结果。此外，引入 Conditional MoEs 仍然保持了通用模型在新任务上进行零样本推理的泛化能力，*e.g.,* 视频-文本检索和视频字幕生成。代码和预训练通用模型将在 https://github.com/fundamentalvision/Uni-Perceiver 发布。

1 引言

长期以来，机器学习领域一直在追求能够处理多种模态和众多任务的通用模型。然而，以往的研究 [65, 89, 71] 侧重于开发带有任务特定模块的专用模型。当这些模型应用于新任务时，需要按需重新设计这些专门设计的组件，并在足够的下游数据上进行微调。因此，它们的模型规模会随着多样化下游任务数量的增加而增长，这与通用模型的目标相冲突。

最近，一些先驱研究 [93, 79, 3, 84, 86, 62] 已经初步尝试通过将各种任务建模为统一形式来构建通用模型。借助统一建模，在各种数据集上进行大规模预训练使通用模型能够使用共享参数处理不同的下游任务。这些通用模型不仅在预训练任务上取得了具有竞争力的性能 [79, 3, 84, 86]，还能够在不引入额外参数的情况下对新任务进行零样本推理 [93, 62]。

然而，与为每个任务配备特定参数的专用模型相比，使用共享参数的通用模型会受到任务干扰问题的影响——不同任务共享

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parameters may conflict with each other [88]. The same issue is also observed in multilingual NLP models [4, 81, 83]. We argue that the task-interference issue is mainly caused by the inconsistent optimization in multi-task learning. As shown in Tab. 1, during the training phase of generalist models, the gradient directions of different tasks would be inconsistent or even opposite. Thus, if multiple tasks share parameters, the optimal update direction of the shared parameters will be uncertain, resulting in sub-optimal performance.

Allowing conflicting modalities and tasks to use separate parameters should effectively mitigate the interference issue in generalist models. Mixture of Experts (MoEs) [43, 23] provides a potential solution, which learns to activate sub-networks dynamically without introducing any task-specific modules. Nevertheless, vanilla MoEs [67] select the experts according to token representations, which suffers from high training/inference cost and neglects the information of different tasks and modalities. In this work, we argue that routing strategies of MoEs require special design when applied to generalist models for mitigating the task-interference issue.

To address the task-interference issue in generalist models, we propose Conditional Mixture-of-Experts (Conditional MoEs), which improve vanilla MoEs by introducing information under different levels of conditions, including token-level, context-level, modality-level, task-level, and predefined token attributes. In this case, vanilla MoEs is a token-level variant of our Conditional MoEs, which can be replaced by other-level variants to implement stronger generalist models. We carefully discussed the training/inference cost and generalization ability of different variants, and ablated their performances in mitigating the interference issue of generalist models. Notably, Conditional MoEs with predefined token attributes introduces 8-bit attribute embedding to describe the information of currently processed task and modalities, which demonstrate excellent computational and memory efficiency and good generalization ability.

To verify the effectiveness of Conditional MoEs, we incorporated it with the recently proposed generic perception model Uni-Perceiver [93] by replacing the linear projection in self-attention and FFN blocks with conditional MoE layers. Experiments demonstrate that, by mitigating task interference with our proposed Conditional MoEs, Uni-Perceiver can be pre-trained on various tasks jointly without performance degradation, while its generalization to other tasks can be maintained simultaneously. Our main contributions are as follows:

- We carefully analyze the task-interference issue in generalist models, and provide an explanation from the gradient direction perspective as well as a metric to quantify the issue.
- We propose Conditional MoEs to address the task-interference issue in generalist models. By introducing the information of currently processed task and modalities, Conditional MoEs effectively mitigate the interference issue, while keeping low computational and memory cost.
- Compared with previous SOTAs, our generalist model with 1% downstream data prompt tuning achieves competitive performance, while only $<$5% training data and $<$10% training cost are used. We hope this work can serve as a solid baseline for generalist models and motivate further research.

2 Related Works

Specialized Models. Previous research focuses on building specialized models for specific tasks. CNNs [47, 26, 70] and ViTs [20, 53, 76, 80] are developed for image classification. Subsequent works re-design them to adapt to diverse downstream visual tasks, *e.g.,* object detection [63] and segmentation [15, 48]. In NLP, different architectures are specifically designed for neural machine translation [77], natural language understanding [19], and natural language generation [51]. As for vision-language tasks, previous works usually combined modality-specific encoders and representation fusion modules together [13, 54]. Recently, [89, 65, 71] integrate several specialized models into a single one to handle diverse tasks. Such integrated specialized models are equipped with multiple task-specific modules to adapt to as many downstream tasks as possible. However, these methods still follow the task-specific paradigm, which conflicts with the objective of generalist models.

Vanilla Generalist Models. Vanilla generalist models handle different tasks and modalities with shared parameters. Uni-Perceiver [93] formulates various perception tasks as finding the maximum likelihood target for each input through the similarity of their representations. OFA [79], Flamingo [3] and SimVLM [84] attempt to unify different tasks into sequence-to-sequence generation. UniCORN [86] and Gato [62] further incorporate bounding box and reinforcement learning tasks into the unified formulation, respectively. These generalist models not only achieve competitive

参数之间可能会相互冲突 [88]。在多语言 NLP 模型中也观察到了同样的问题 [4, 81, 83]。我们认为，任务干扰问题主要是由多任务学习中的优化不一致导致的。如表 1 所示，在通用模型的训练阶段，不同任务的梯度方向可能不一致，甚至相反。因此，如果多个任务共享参数，共享参数的最优更新方向将变得不确定，从而导致次优性能。

允许存在冲突的模态和任务使用独立参数，应能有效缓解通用模型中的干扰问题。专家混合（Mixture of Experts, MoEs）[43, 23] 提供了一种潜在解决方案，其能够在不引入任何任务特定模块的情况下动态激活子网络。然而，原始 MoEs [67] 根据 token 表示来选择专家，这不仅会带来较高的训练/推理成本，还会忽略不同任务和模态的信息。在本文中，我们认为，当 MoEs 应用于通用模型以缓解任务干扰问题时，其路由策略需要进行专门设计。

为了解决通用模型中的任务干扰问题，我们提出了条件混合专家（Conditional Mixture-of-Experts，Conditional MoEs），通过引入不同条件层级的信息来改进普通 MoEs，包括词元级、上下文级、模态级、任务级以及预定义词元属性。在这种情况下，普通 MoEs 是我们 Conditional MoEs 的一种词元级变体，可以由其他层级的变体替代，以实现更强大的通用模型。我们仔细讨论了不同变体的训练/推理成本和泛化能力，并通过消融实验评估了它们在缓解通用模型干扰问题方面的性能。值得注意的是，带有预定义词元属性的 Conditional MoEs 引入了 8 位属性嵌入，用于描述当前处理任务和模态的信息，展现出优异的计算和内存效率以及良好的泛化能力。

为了验证条件 MoE 的有效性，我们将其与近期提出的通用感知模型 Uni-Perceiver [93] 相结合，将自注意力和 FFN 模块中的线性投影替换为条件 MoE 层。实验表明，通过使用我们提出的条件 MoE 缓解任务干扰，Uni-Perceiver 可以在不降低性能的情况下对多种任务进行联合预训练，同时还能保持其对其他任务的泛化能力。我们的主要贡献如下：

- 我们仔细分析了通用模型中的任务干扰问题，并从梯度方向的角度提供了解释，同时提出了一个用于量化该问题的指标。
- 我们提出条件 MoE，以解决通用模型中的任务干扰问题。通过引入当前处理任务和模态的信息，条件 MoE 能够有效缓解干扰问题，同时保持较低的计算和内存成本。
- 与以往的 SOTA 相比，我们的通用模型仅使用 1% 的下游数据进行提示调优，便取得了有竞争力的性能，同时仅使用了 <5% 的训练数据和 <10% 的训练成本。我们希望这项工作能够作为通用模型的坚实基线，并推动进一步研究。

2 相关工作

专用模型。以往研究专注于为特定任务构建专用模型。CNN [47, 26, 70] 和 ViT [20, 53, 76, 80] 被开发用于图像分类。后续工作对它们进行重新设计，以适应多样化的下游视觉任务，*e.g.,* 如目标检测 [63] 和分割 [15, 48]。在自然语言处理领域，不同架构被专门设计用于神经机器翻译 [77]、自然语言理解 [19] 和自然语言生成 [51]。至于视觉-语言任务，以往工作通常将特定模态的编码器与表示融合模块结合起来 [13, 54]。近期，[89, 65, 71] 将多个专用模型整合为一个模型，以处理多样化任务。这类集成式专用模型配备了多个任务特定模块，以尽可能适应更多下游任务。然而，这些方法仍然遵循任务特定范式，这与通用模型的目标相冲突。

原生通用模型。原生通用模型使用共享参数处理不同任务和模态。Uni-Perceiver [93] 将各种感知任务表述为：通过表示之间的相似性，为每个输入寻找最大似然目标。OFA [79]、Flamingo [3] 和 SimVLM [84] 试图将不同任务统一为序列到序列生成。UniCORN [86] 和 Gato [62] 分别进一步将边界框和强化学习任务纳入统一表述中。这些通用模型不仅取得了具有竞争力的

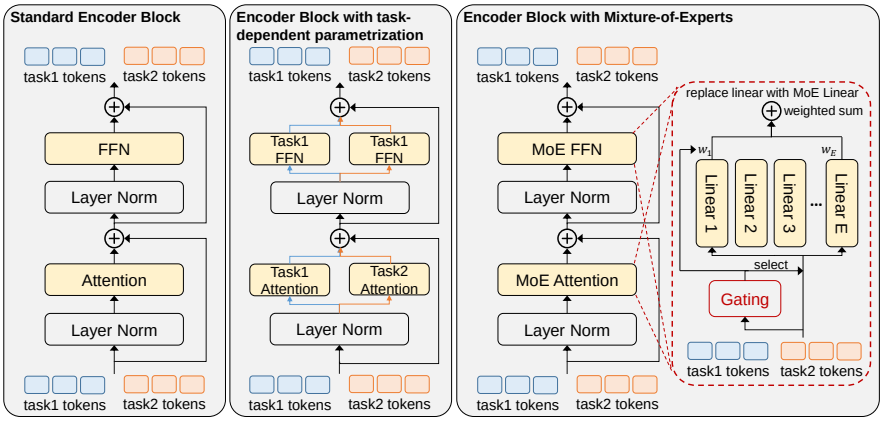


Figure 1: Comparisons of fully-shared standard encoder block, task-specific encoder block with task-dedicated parameters, and encoder block with efficient MoE parameterization.

performance on pre-training tasks with shared parameters, but also can perform zero-shot inference on new tasks [62, 93]. However, these methods rarely investigate the potential interference among different modalities and tasks, which could result in the performance degradation of generalist models.

Multi-Task Learning. Multi-task learning [8, 17] has been widely studied in the community of vision [27, 74, 72], language [25, 16, 50] and vision-language learning [10, 55, 29]. While multi-task training enables collaboration between tasks, it may also introduce the task interference problem [81, 83, 28, 36, 72]. To mitigate the task-interference issue, some works[14, 24, 38] propose to dynamically adjust the loss weight for each task, while others [90, 49, 36] instead use task-dedicated parameters. However, methods with task-specific parameters are difficult to generalize to new tasks and do not meet the requirements of generalist models.

Mixture of Experts (MoEs). MoEs has shown its remarkable ability to scale neural networks [67, 43, 23, 64, 21]. [67] first proves the effectiveness of MoEs by stacking MoE layers in the LSTM models. [68, 43] further introduce this approach to Transformer architectures. [23, 40] train language models with trillion parameters successfully by utilizing simplified MoE routing strategy and efficient training techniques. There are also some works applying MoEs to CNNs for computer vision tasks [1, 85, 82]. Recently, V-MoE [64] successfully employs MoEs to ViTs, showing promising performance on many visual tasks. Task-MoE [42] focuses on applying MoEs for multilingual translation to mitigate the interference among different languages. In this work, we aim to explore MoEs under different conditions for general models.

3 Methodology

In this section, we first analyze the task-interference problem from the gradient direction perspective. Based on the analysis, we propose Conditional Mixture-of-Experts (Conditional MoEs) for generalist models, which introduces parameters conditioned by information of different levels to mitigate the task-interference issue with negligible overhead.

3.1 Task Interference

To quantify the interference of the j-th task on the i-th task, we estimate the change in loss L_i of the i-th task, when optimizing the shared parameters θ according to the j-th task L_j as:

$$\Delta_j L_i(x_i) \doteq \mathbb{E}_{x_j}\left(L_i(x_i;\theta) - L_i(x_i;\theta - \lambda\frac{\nabla_\theta L_j(x_j)}{\|\nabla_\theta L_j(x_j)\|})\right) \approx \lambda \mathbb{E}_{x_j}\left(\frac{\nabla_\theta L_j(x_j)}{\|\nabla_\theta L_j(x_j)\|}^T \nabla_\theta L_i(x_i)\right), \quad (1)$$

where x_i and x_j are the sampled training batches of the i-th and j-th tasks, respectively. Without loss of generality, we only consider the update direction ignoring the update norm. Then, the interference of the j-th task on the i-th task can be quantified as:

$$\mathcal{I}_{i,j} = \mathbb{E}_{x_i}\left(\frac{\Delta_j L_i(x_i)}{\Delta_i L_i(x_i)}\right), \quad (2)$$

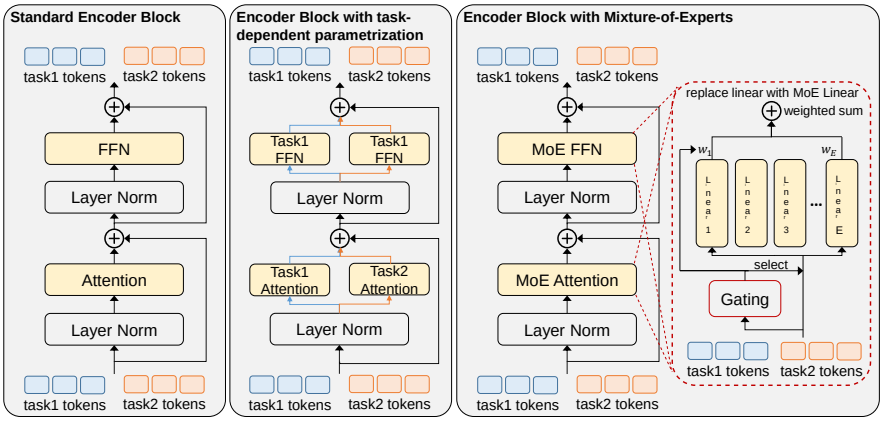


图 1：全共享标准编码器块、具有任务专用参数的任务特定编码器块以及采用高效 MoE 参数化的编码器块的比较。

在使用共享参数进行预训练任务时的性能，而且还可以在新任务上进行零样本推理 [62, 93]。然而，这些方法很少研究不同模态和任务之间潜在的干扰，而这种干扰可能导致通用模型的性能下降。

多任务学习。多任务学习 [8, 17] 已在视觉 [27, 74, 72]、语言 [25, 16, 50] 以及视觉-语言学习 [10, 55, 29] 领域得到广泛研究。尽管多任务训练能够促进任务之间的协作，但也可能引入任务干扰问题 [81, 83, 28, 36, 72]。为缓解任务干扰问题，一些工作 [14, 24, 38] 提出动态调整每个任务的损失权重，而另一些工作 [90, 49, 36] 则使用任务专用参数。然而，采用任务特定参数的方法难以泛化到新任务，也不符合通用模型的要求。

专家混合（MoEs）。MoEs 已展现出其扩展神经网络的卓越能力 [67, 43, 23, 64, 21]。[67] 首次通过在 LSTM 模型中堆叠 MoE 层证明了 MoEs 的有效性。[68, 43] 进一步将这一方法引入 Transformer 架构。[23, 40] 通过采用简化的 MoE 路由策略和高效训练技术，成功训练了具有万亿参数的语言模型。也有一些工作将 MoEs 应用于 CNN，以完成计算机视觉任务 [1, 85, 82]。最近，V-MoE [64] 成功地将 MoEs 应用于 ViTs，在许多视觉任务上展现出有前景的性能。Task-MoE [42] 专注于将 MoEs 应用于多语言翻译，以缓解不同语言之间的干扰。在本工作中，我们旨在探索通用模型在不同条件下的 MoEs。

3 方法

在本节中，我们首先从梯度方向的角度分析任务干扰问题。基于该分析，我们提出了面向通用模型的条件混合专家（Conditional MoEs），其引入由不同层级信息条件化的参数，以在几乎不增加额外开销的情况下缓解任务干扰问题。

3.1 任务干扰

为了量化第 j 个任务对第 i 个任务的干扰，我们估计当根据第 j 个任务 L_j 优化共享参数 θ 时，第 i 个任务的损失 L_i 的变化，如下所示：

$$\Delta_j L_i(x_i) \doteq \mathbb{E}_{x_j}\left(L_i(x_i;\theta) - L_i(x_i;\theta - \lambda\frac{\nabla_\theta L_j(x_j)}{\|\nabla_\theta L_j(x_j)\|})\right) \approx \lambda\mathbb{E}_{x_j}\left(\frac{\nabla_\theta L_j(x_j)}{\|\nabla_\theta L_j(x_j)\|}^T \nabla_\theta L_i(x_i)\right), \quad (1)$$

其中，x_i 和 x_j 分别是第 i 个任务和第 j 个任务的采样训练批次。不失一般性，我们仅考虑更新方向，而忽略更新范数。那么，第 j 个任务对第 i 个任务的干扰可以量化为：

$$\mathcal{I}_{i,j} = \mathbb{E}_{x_i}\left(\frac{\Delta_j L_i(x_i)}{\Delta_i L_i(x_i)}\right), \quad (2)$$

Table 1: The average interference metric $\mathcal{I}_{i,j}$ of the task j on the task i at the 4-th/12-nd FFN blocks. To calculate the interference metric, we sample 100 batches for each tasks, and record the gradients based on the pre-trained Uni-Perceiver-Ti. The red value indicates that the task j has a negative impact on the task i, and the green value indicates a positive impact.

(a) The 4-th FFN Block

Task i \ Task j	ImgCLS (Img)	MLM (Text)	Caption (Img-Text)
ImgCLS (Img)	1.00	-0.57	1.29
MLM (Text)	0.07	1.00	0.68
Caption (Img-Text)	0.01	0.01	1.00

(b) The 12-nd FFN Block

Task i \ Task j	ImgCLS (Img)	MLM (Text)	Caption (Img-Text)
ImgCLS (Img)	1.00	-2.91	-2.45
MLM (Text)	-1.65	1.00	-1.05
Caption (Img-Text)	-0.11	0.19	1.00

where the denominator is used to normalize the loss change scale. As reported in Tab. 1, we sample 100 batches for each tasks, and record the gradients to calculate the average interference metric $\mathcal{I}_{i,j}$ of the j-th task on the i-th task at the 4-th/12-nd FFN blocks. We see that, at shallow layers, the image caption task has positive impacts on image classification and masked language modeling, suggesting that cooperation between different tasks exists. While at deep layers, tasks with different optimization objectives hardly enhance each other, and the gradient directions may even opposite.

Fig. 1 summarizes three mainstream architectures for multi-task models. The first is the standard architecture [93, 31, 32] with parameters fully shared by different tasks, which suffers from task interference problem as analyzed above. The second is task-specific parameterized architecture [89, 65, 29, 71] equipped with dedicated parameters for each task. Although this architecture address the interference problem by task-specific parameters, it is difficult to generalize to new tasks that did not emerge in the training phase. Unlike the above two architectures, the Mixture-of-Experts (MoE) architecture [67, 43, 23, 40, 64] activates models sparsely according to different given inputs by selectively utilizing different subset of the model parameters. The sparse routing mechanism makes it possible to train very large generalist models, which maximizes the collaboration and meanwhile mitigates the interference problem. In this work, we focus on exploring Conditional MoEs for general models, whose experts are gated by conditions from different levels.

3.2 Conditional Mixture-of-Experts (Conditional MoEs)

We first describe the prototype of Conditional MoEs, and then provide its specific instantiations under different conditions, as well as the application to generalist models.

Prototype. Given any token x_i in the input sequence $X = \{x_i\}_{i=1}^{L}$, conditional MoEs with E experts firstly introduces a gate decision vector $\mathcal{G} \in \mathbb{R}^E$ that dispatches different input tokens to different experts, which is calculated as:

$$\mathcal{G} = \text{top}_k\left(\text{softmax}\left(\mathbf{W}_g \cdot R(x_i) + \epsilon\right)\right). \tag{3}$$

where $R(\cdot)$ defines a general routing strategy for gate decision, which is alternative under different conditions. $\mathbf{W}_g$ is the trainable weights in gate decision and ϵ is the noise term. The $\text{top}_k(\cdot)$ operator sets all values to be zero except the largest k values. Since $\mathcal{G}$ only has $k \ll E$ non-zero values, the token x_i is routed to only a small number of experts. After getting the gate decision vector $\mathcal{G}$, the corresponding output y_i is the weighted combination of each expert's computation on x_i as:

$$y_i = \sum_{e=1}^{E} \mathcal{G}_e \cdot \mathbf{W}_e \cdot x_i, \tag{4}$$

where $\mathbf{W_e}$ is the linear projection weights of the e-th expert and gate decision $\mathcal{G}_e$ determines how much the e-th expert contributes to the output y_i. Note that, experts with $\mathcal{G}_e = 0$ does not need to be computed for saving computation.

In Conditional MoEs, the routing strategy $R(\cdot)$ plays an important role in the multi-modality and multi-task training of generalist models. By sparsely activating experts according to different conditions, Conditional MoEs can mitigate the interference issue while maintaining the generality of the pretrained model. Next, we introduce variants with specific routing strategies under different conditions, as shown in Fig. 2.

Token-Level Routing. Similar to vanilla MoEs [67, 43, 23, 40, 64], the token-Level MoEs directly use the token representation for the routing strategy, which can be written as:

$$R_{\text{token}}(x_i) = x_i. \tag{5}$$

表 1：任务 j 对任务 i 在第 4/12 个 FFN 块处的平均干扰指标 $\mathcal{I}_{i,j}$。为计算干扰指标，我们为每个任务采样 100 个批次，并基于预训练的 Uni-Perceriver-Ti 记录梯度。红色数值表示任务 j 对任务 i 产生负面影响，绿色数值表示产生正面影响。

(a) The 4-th FFN Block

Task i \ Task j	ImgCLS (Img)	MLM (Text)	Caption (Img-Text)
ImgCLS (Img)	1.00	-0.57	1.29
MLM (Text)	0.07	1.00	0.68
Caption (Img-Text)	0.01	0.01	1.00

(b) The 12-nd FFN Block

Task i \ Task j	ImgCLS (Img)	MLM (Text)	Caption (Img-Text)
ImgCLS (Img)	1.00	-2.91	-2.45
MLM (Text)	-1.65	1.00	-1.05
Caption (Img-Text)	-0.11	0.19	1.00

其中，分母用于归一化损失变化的尺度。如表 1 所示，我们为每个任务采样 100 个批次，并记录梯度，以计算第 j 个任务在第 4/第 12 个 FFN 块处对第 i 个任务的平均干扰指标 $\mathcal{I}_{i,j}$。可以看到，在浅层中，图像描述任务对图像分类和掩码语言建模具有正向影响，表明不同任务之间存在协作。而在深层中，具有不同优化目标的任务几乎无法相互增强，梯度方向甚至可能相反。

图 1 总结了多任务模型的三种主流架构。第一种是标准架构 [93, 31, 32]，不同任务完全共享参数，如上文分析所示，这种架构存在任务干扰问题。第二种是任务特定参数化架构 [89, 65, 29, 71]，为每个任务配备专用参数。尽管该架构通过任务特定参数解决了干扰问题，但它难以泛化到训练阶段未出现的新任务。不同于上述两种架构，混合专家（Mixture-of-Experts，MoE）架构 [67, 43, 23, 40, 64] 会根据不同给定输入，通过选择性地利用模型参数的不同子集来稀疏地激活模型。稀疏路由机制使训练超大规模通用模型成为可能，从而最大化协作，同时缓解干扰问题。在本文中，我们重点探索用于通用模型的条件 MoE，其专家由来自不同层级的条件进行门控。

3.2 条件混合专家（Conditional MoEs）

我们首先描述条件式 MoE 的原型，然后给出其在不同条件下的具体实例化方式，以及其在通用模型中的应用。

原型。给定输入序列 $X = \{x_i\}_{i=1}^{L}$ 中的任意 token x_i，具有 E 个专家的条件式 MoE 首先引入一个门控决策向量 $\mathcal{G} \in \mathbb{R}^E$，用于将不同的输入 token 分派给不同的专家，其计算方式如下：

$$\mathcal{G} = \text{top}_k\left(\text{softmax}\left(\mathbf{W}_g \cdot R(x_i) + \epsilon\right)\right). \tag{3}$$

其中，$R(\cdot)$ 定义了一种用于门控决策的通用路由策略，该策略在不同条件下具有不同选择。$\mathbf{W}_g$ 是门控决策中的可训练权重，ϵ 是噪声项。$\text{top}_k(\cdot)$ 算子将除最大的 k 个值之外的所有值置为零。由于 $\mathcal{G}$ 只有 $k \ll E$ 个非零值，令牌 x_i 仅被路由到少数专家。在得到门控决策向量 $\mathcal{G}$ 后，相应的输出 y_i 是每个专家在 x_i 上计算结果的加权组合，如下所示：

$$y_i = \sum_{e=1}^{E} \mathcal{G}_e \cdot \mathbf{W}_e \cdot x_i, \tag{4}$$

其中，$\mathbf{W_e}$ 是第 e 个专家的线性投影权重，门控决策 $\mathcal{G}_e$ 决定第 e 个专家对输出 y_i 的贡献程度。注意，$\mathcal{G}_e =$ 为 0 的专家无需计算，以节省计算量。

在条件式 MoE 中，路由策略 $R(\cdot)$ 在通用模型的多模态和多任务训练中发挥着重要作用。通过根据不同条件稀疏激活专家，条件式 MoE 可以在保持预训练模型通用性的同时缓解干扰问题。接下来，我们介绍在不同条件下采用特定路由策略的变体，如图 2 所示。

词元级路由。与普通 MoE [67, 43, 23, 40, 64] 类似，词元级 MoE 直接使用词元表示来制定路由策略，可写为：

$$R_{\text{token}}(x_i) = x_i. \tag{5}$$

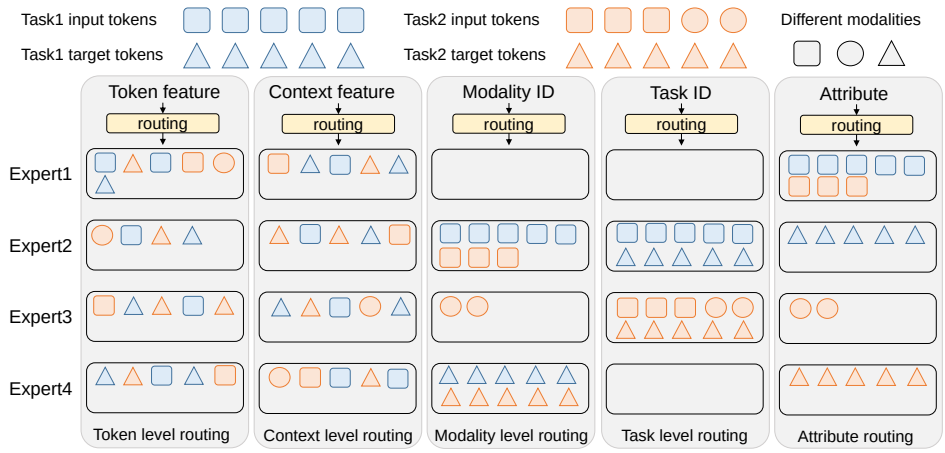


Figure 2: Comparisons of routing strategies with the top-1 gate decisions under 2-task training.

The routing strategy of token-level MoEs is an identical function, where the gate decision only depends on each token's own representation.

Context-Level Routing. Tokens with similar representations may appear in conflicting tasks, whose optimal expert decisions should be different to mitigate the task interference. Therefore, to help gate function making more reliable decisions, we explore the combination of global context and local token representation. The routing strategy utilizing global context can be expressed as:

$$R_{\text{context}}(x_i) = \text{concat}(x_i, \text{attnpool}(X)), \tag{6}$$

where $\text{concat}(\cdot)$ indicates the concatenation operation, $X = \{x_i\}_{i=1}^{L}$ is sequence of all tokens in the current sample, and $\text{attnpool}(\cdot)$ indicates the attention pooling operator [61].

Modality-Level Routing. Most current practice uses modality-specific encoders with independent parameters for different modality inputs. Inspired by this, we also explore to leverage the modality of the current token as a routing strategy:

$$R_{\text{modal}}(x_i) = \text{embed}(\text{id}_{\text{modal}}(x_i)) \tag{7}$$

Here, $\text{embed}(\cdot)$ represent the embedding layer, and $\text{id}_{\text{modal}}(\cdot)$ indicates the modality index of current token x_i. The routing function will assign this token to experts according to its modality embedding.

Task-Level Routing. In addition to modality information, task information can also be used to guide gate functions to make reliable decisions for mitigating the task interference. Similar to Eqn. (7), the routing strategy can be formulated as:

$$R_{\text{task}}(x_i) = \text{embed}(\text{id}_{\text{task}}(x_i)), \tag{8}$$

where $\text{id}_{\text{task}}(\cdot)$ is the task index of current token x_i. The task embedding for this task will be used to compute gate decision. Since all tokens from one task have the same task embedding, all tokens corresponding to this task will be routed to the same set of experts.

Attribute Routing. Among the aforementioned variants, token-level, context-level, and modality-level routing strategies only focus on input tokens but omit information about the currently processed task. While task-level routing strategy relies on task-specific ids, it limits the generalization ability to new downstream tasks. To introduce the information of currently processed task and modalities without losing the generalization ability of generalist models, we propose to introduce token attributes to assist the gate decision.

As described in Tab. 2, the attributes of the current token are represented as an 8-dimensional binary embedding, whose attributes include the modalities of current task and token (index 0∼5), the causation type of the model (index 6), and the token source (index 7). As a result, the designed token attributes provide comprehensive information of currently processed task meanwhile keeping the task generalization ability. Based on the attribute embedding, the routing strategy is expressed as:

$$R_{\text{attr}}(x_i) = \text{layernorm}\left(\mathbf{W}_{\text{attr}} \cdot \text{attr}(x_i)\right). \tag{9}$$

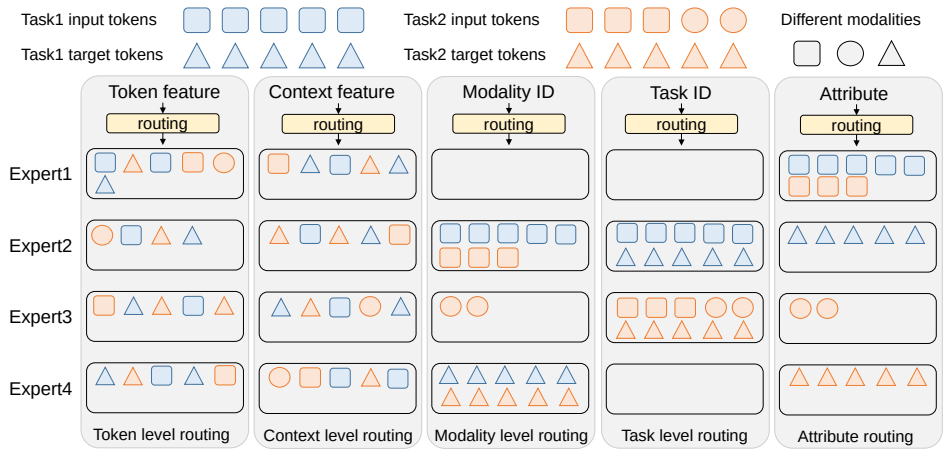


图 2：在双任务训练下，采用 top-1 门控决策的路由策略比较。

词元级 MoE 的路由策略是一个恒等函数，其中门控决策仅取决于每个词元自身的表示。

上下文级路由。具有相似表示的词元可能出现在相互冲突的任务中，其最优专家决策应当有所不同，以缓解任务干扰。因此，为了帮助门控函数做出更可靠的决策，我们探索全局上下文与局部词元表示的结合。利用全局上下文的路由策略可以表示为：

$$R_{\text{context}}(x_i) = \text{concat}(x_i, \text{attnpool}(X)), \tag{6}$$

其中 concat(·) 表示拼接操作，$X = \{x_i\}_{i=1}^{L}$ 是当前样本中所有词元的序列，attnpool(·) 表示注意力池化算子 [61]。

模态级路由。当前大多数实践针对不同模态输入使用具有独立参数的模态特定编码器。受此启发，我们也探索利用当前 token 的模态作为一种路由策略：

$$R_{\text{modal}}(x_i) = \text{embed}(\text{id}_{\text{modal}}(x_i)) \tag{7}$$

这里，embed(·) 表示嵌入层，$\text{id}_{\text{modal}}(\cdot)$ 表示当前令牌 x_i 的模态索引。路由函数将根据该令牌的模态嵌入将其分配给专家。

任务级路由。除了模态信息外，任务信息也可用于引导门控函数做出可靠决策，以缓解任务干扰。与公式（7）类似，路由策略可表示为：

$$R_{\text{task}}(x_i) = \text{embed}(\text{id}_{\text{task}}(x_i)), \tag{8}$$

其中 $\text{id}_{\text{task}}(\cdot)$ 是当前 token x_i 的任务索引。该任务的任务嵌入将用于计算门控决策。由于来自同一任务的所有 token 都具有相同的任务嵌入，因此对应于该任务的所有 token 都将被路由到同一组专家。

属性路由。在上述变体中，词元级、上下文级和模态级路由策略仅关注输入词元，而忽略了当前所处理任务的信息。尽管任务级路由策略依赖于特定任务的 ID，但这会限制其对新的下游任务的泛化能力。为在不损害通用模型泛化能力的情况下引入当前所处理任务和模态的信息，我们提出引入词元属性来辅助门控决策。

如表 2 所述，当前 token 的属性被表示为一个 8 维二进制嵌入，其属性包括当前任务和 token 的模态（索引 0~5）、模型的因果类型（索引 6）以及 token 来源（索引 7）。因此，所设计的 token 属性在保持任务泛化能力的同时，为当前处理的任务提供了全面的信息。基于属性嵌入，路由策略表示为：

$$R_{\text{attr}}(x_i) = \text{layernorm}\left(\mathbf{W}_{\text{attr}} \cdot \text{attr}(x_i)\right). \tag{9}$$

Table 2: The 8-dimensional binary embedding used for attribute-level routing strategy. The attribute embedding is assigned to a token by checking whether the statements of the eight descriptions match the current token. For example, the attribute embedding for any token from the input sequences of image classification task should be $[1, 0, 0, 1, 1, 0, 0, 1]$. Please refer to the Appendix for detailed look-up table of attribute embeddings for all tasks in our work.

Index	Descriptions	Yes	No
0	Visual modality exists in the inputs of the current task.	1	0
1	Text modality exists in the inputs of the current task.	1	0
2	Visual modality exists in the targets of the current task.	1	0
3	Text modality exists in the targets of the current task.	1	0
4	The modality of current token is visual.	1	0
5	The modality of current token is text.	1	0
6	The attention mask of the current token is causal.	1	0
7	The current token comes from the inputs, not the targets.	1	0

Here, $\mathrm{attr}(x_i)$ is the 8-dimensional binary attribute embedding of the current token x_i as described in Tab. 2. $\mathbf{W}_{\text{attr}}$ is the learnable weights to transform the attribute embedding to latent representation, and $\mathrm{layernorm}(\cdot)$ denotes the layer normalization [6] for training stabilization.

Application to Generalist Models. Without loss of generality, we explore the application of Conditional MoEs to the generalist model Uni-Perceiver [93], which uses Transformers to handle various modalities and tasks with shared parameters. We replace linear projection layers in both self-attention and FFN blocks with Conditional-MoE layers (see Fig. 1).

3.3 Comparison of Conditional-MoE Variants

As illustrated in Fig. 2, among the variants of Conditional MoEs, token-level and context-level MoEs are data-dependent, while modality-level, task-level, and attribute MoEs are data-independent.

Training and Inference Cost. Compared to dense models with the same number of parameters, all Conditional MoE variants can significantly reduce the computational cost benefiting from the sparse routing mechanism. Due to the dependence of input data, the memory consumption of token-level and context-level MoEs is relatively high during model training, and model parallelism is required to relieve memory cost by partitioning experts across multiple devices, leading to heavy inter-device communication overhead. This problem persists when using pre-trained models for task-specific inference, where all experts need to be loaded into memory and might be activated by any token.

Different from data-dependent Conditional MoEs, data-independent variants such as modality-level, task-level, and attribute MoEs have excellent memory efficiency, since only top-k experts need to be activated for all tokens with the same modality/task/attributes. Moreover, in both training and inference phase, the experts in a data-independent MoE layer can be merged into a single linear projection using reparameterization techniques. In this case, the computation cost of the network with data-independent Conditional MoEs will be equivalent to a dense model without MoEs.

Generalization Ability. We hope to mitigate the task-interference issue in generalist models, while keeping their generalization ability to new downstream tasks. While token-level, context-level, and modality-level MoEs without task-specific designs do not harm the generalization ability, they ignore the task-level information which is essential to resolve the task interference. Conversely, task-level routing strategy is tied to a specific task id, which is difficult to generalize to new downstream tasks. Attribute MoEs introduce predefined token attributes to comprehensively describe the information of currently processed task and modalities, which can be transferred to new downstream tasks without any task-specific modifications. This gives attribute MoEs the potential to mitigate task interference without losing generalization ability.

4 Experiments

In this section, we first describe our experimental setup. Then, we confirm the task-interference issue in the generalist model Uni-Perceiver [93] and ablate the the ability of different Conditional MoEs to mitigate task interference. Finally, large-scale training is conducted to verify the effectiveness of our proposed Conditional-MoEs and its generalization ability to novel tasks.

表2：用于属性级路由策略的8维二进制嵌入。通过检查八个描述的陈述是否与当前标记匹配，将属性嵌入分配给一个标记。例如，图像分类任务输入序列中任意标记的属性嵌入应为 [1, 0, 0, 1, 1, 0, 0, 1]。有关我们工作中所有任务的属性嵌入的详细查找表，请参见附录。

Index	Descriptions	Yes	No
0	Visual modality exists in the inputs of the current task.	1	0
1	Text modality exists in the inputs of the current task.	1	0
2	Visual modality exists in the targets of the current task.	1	0
3	Text modality exists in the targets of the current task.	1	0
4	The modality of current token is visual.	1	0
5	The modality of current token is text.	1	0
6	The attention mask of the current token is causal.	1	0
7	The current token comes from the inputs, not the targets.	1	0

这里，attr(x_i) 是当前标记 x_i 的 8 维二进制属性嵌入，如表 2 所述。$\mathbf{W}_{\text{attr}}$ 是用于将属性嵌入转换为潜在表示的可学习权重，而 layernorm(·) 表示用于稳定训练的层归一化 [6]。

应用于通用模型。在不失一般性的情况下，我们探索了条件 MoE 在通用模型 Uni-Perceiver [93] 中的应用，该模型使用 Transformer 通过共享参数处理各种模态和任务。我们将自注意力和 FFN 模块中的线性投影层替换为条件 MoE 层（见图 1）。

3.3 Conditional-MoE 变体的比较

如图 2 所示，在条件 MoE 的各种变体中，token 级和上下文级 MoE 是数据依赖的，而模态级、任务级和属性 MoE 是数据无关的。

训练和推理成本。与参数数量相同的稠密模型相比，所有条件 MoE 变体都能受益于稀疏路由机制，从而显著降低计算成本。由于对输入数据的依赖，token 级和上下文级 MoE 在模型训练期间的内存消耗相对较高，需要通过将专家划分到多个设备上来进行模型并行，以缓解内存成本，但这会导致沉重的设备间通信开销。在使用预训练模型进行特定任务推理时，这一问题依然存在，因为所有专家都需要加载到内存中，并且可能被任意 token 激活。

不同于数据依赖的条件 MoE，模态级、任务级和属性 MoE 等数据无关变体具有出色的内存效率，因为对于具有相同模态/任务/属性的所有 token，只需激活 top-k 个专家。此外，在训练和推理阶段，数据无关 MoE 层中的专家都可以通过重参数化技术合并为单个线性投影。在这种情况下，带有数据无关条件 MoE 的网络的计算成本将等同于不含 MoE 的稠密模型。

泛化能力。我们希望在保持通用模型对新的下游任务的泛化能力的同时，缓解任务干扰问题。尽管没有任务特定设计的词元级、上下文级和模态级 MoE 不会损害泛化能力，但它们忽略了对解决任务干扰至关重要的任务级信息。相反，任务级路由策略与特定的任务 ID 绑定，难以泛化到新的下游任务。属性 MoE 引入预定义的词元属性，以全面描述当前处理任务和模态的信息，并且无需任何任务特定的修改即可迁移到新的下游任务。这使得属性 MoE 有潜力在不丧失泛化能力的情况下缓解任务干扰。

4 实验

在本节中，我们首先描述实验设置。然后，我们确认通用模型 Uni-Perceiver [93] 中存在的任务干扰问题，并对不同 Conditional MoE 缓解任务干扰的能力进行消融分析。最后，我们进行大规模训练，以验证我们提出的 Conditional-MoEs 的有效性及其对新任务的泛化能力。

Table 3: The performance of different routing strategies for Conditional MoEs. The base model is Uni-Perceiver with $\text{BERT}_{\text{tiny}}$. We also illustrate the task-specific variant where each task has its own specialized parameters. The training and validation performance reported on three tasks: image classification on ImageNet-1K [18], image caption on COCO Caption [12], and Masked Language Modeling(MLM) on Books&Wiki. The best results within a tolerance of 1% are in **bold**.

model	task-specific parameterization	training time	inference time	ImageNet-1k		COCO Caption		MLM	
				$\uparrow\text{acc}_{\text{train}}$	$\uparrow\text{acc}_{\text{val}}$	$\uparrow\text{acc}_{\text{train}}$	$\uparrow\text{B@4}_{\text{val}}$	$\uparrow\text{acc}_{\text{train}}$	$\downarrow\text{ppl}_{\text{val}}$
Uni-Perceiver-Ti [93]		1.0×	1.0×	47.3	68.3	49.2	18.2	54.5	5.86
	✓	1.1×	1.0×	**53.3**	**73.5**	**52.6**	20.4	**60.5**	**4.48**
+ Conditional MoEs $_{\text{token}}$		1.8×	2.2×	**53.1**	72.7	**52.9**	20.9	58.3	4.96
+ Conditional MoEs $_{\text{context}}$		2.2×	2.6×	52.5	**73.1**	**52.8**	21.5	58.6	4.86
+ Conditional MoEs $_{\text{modality}}$		1.4×	1.0×	51.7	72.6	52.1	21.8	57.5	5.06
+ Conditional MoEs $_{\text{task}}$		1.4×	1.0×	**52.9**	**73.2**	**52.7**	21.2	**59.9**	**4.56**
+ Conditional MoEs $_{\text{attribute}}$		1.4×	1.0×	**52.8**	**73.3**	**53.1**	**23.0**	**60.0**	**4.56**

4.1 Datasets

We use the same datasets in Uni-Perceiver [93] to pre-train our models. Specifically, ImageNet-21k [18] is used for image classification pre-training. Kinetics-700 [37] and Moments in Time [57] are used for video classification pre-training. Language modeling task is trained on BookCorpus [94] & English Wikipedia (Books&Wiki). For language modeling with image clues and image-text retrieval, we use a combination of image-text-pair datasets: SBU Captions (SBU) [58], Visual Genome [41], COCO Caption [12], CC3M [66], CC12M [9] and YFCC [35]. Following Uni-Perceiver, Imagenet1K [18], Kinetics-400 [37], COCO Caption [12], and Flickr30k [59] are utilized to evaluate the performance of generalist models on downstream tasks. We also use two datasets that evaluate the generalization ability to novel tasks: MSVD [11] and GLUE [78]. Additionally, all dataset licenses are included in Appendix.

4.2 Implementation Details

We incorporate the vanilla generalist model Uni-Perceiver with Conditional MoEs for experiments with three different variants: Uni-Perceiver-Ti (Tiny), Uni-Perceiver-B (Base), and Uni-Perceiver-L (Large). Please refer to Appendix for architecture hyperparameters. If not specified, the input image resolution is set to 224×224. In each training iteration, each GPU independently samples a single task and dataset. The gradients of different GPUs are synchronized after the gradient back-propagation. We use the AdamW optimizer with a base learning rate of 0.0005 and a weight decay of 0.05. Similar to [52, 61], we find setting $\beta_2 = 0.98$ and $\epsilon = 10^{-6}$ helps improve stability when large-scale training. Besides, gradient clipping with 0.5 is used to stabilize training.

Uni-Perceiver-B and Uni-Perceiver-L are equipped with Conditional-MoEs layer for every other layers while Uni-Perception-Ti use Conditional MoEs in all layers. A normal noise is also used on the gate logits following [64] for a better exploration for new potential experts. If not specialized, top-2 gate function is used. For other hyper-parameters of MoE layers, please refer to Appendix.

4.3 Ablation Studies

This part explores whether Conditional MoEs can effectively mitigate task interference in generalist models and compares different routing strategies. Tab. 3 summarizes the performance of Uni-Perceiver and its variants on three typical tasks. Compared with task-specific parameterization, the performance degradation of Uni-Perceiver confirms the existence of task interference. Incorporating Conditional MoEs with any routing strategy can mitigate the task-interference issue and significantly improve the performance. Among these five routing strategies, token-level, context-level, and modality-level MoEs deliver slightly worse performance. We argue the missed task information is critical for resolving the task interference. Besides, the data-dependent MoEs, *i.e.,* token-level and context-level, have relatively higher training and inference cost, while the other three MoEs have excellent efficiency by using reparameterization techniques. Although both task-level and attribute MoEs achieve good performance, the specialized task-id design in task-level MoEs makes it difficult to generalize to new tasks. Therefore, the Conditional MoEs with attribute routing strategy will be used.

表 3：条件 MoE 不同路由策略的性能。基础模型为采用 $\text{BERT}_{\text{tiny}}$ 的 Uni-Perceiver。我们还展示了任务特定变体，其中每个任务都有自己的专用参数。报告了三个任务上的训练和验证性能：ImageNet-1K [18] 上的图像分类、COCO Caption [12] 上的图像描述，以及 Books&Wiki 上的掩码语言建模（MLM）。在 1% 容差范围内的最佳结果以粗体显示。

model	task-specific parameterization	training time	inference time	ImageNet-1k		COCO Caption		MLM	
				$\uparrow\text{acc}_{\text{train}}$	$\uparrow\text{acc}_{\text{val}}$	$\uparrow\text{acc}_{\text{train}}$	$\uparrow\text{B@4}_{\text{val}}$	$\uparrow\text{acc}_{\text{train}}$	$\downarrow\text{ppl}_{\text{val}}$
Uni-Perceiver-Ti [93]		1.0×	1.0×	47.3	68.3	49.2	18.2	54.5	5.86
	✓	1.1×	1.0×	**53.3**	**73.5**	**52.6**	20.4	**60.5**	**4.48**
+ Conditional MoEs $_{\text{token}}$		1.8×	2.2×	**53.1**	72.7	**52.9**	20.9	58.3	4.96
+ Conditional MoEs $_{\text{context}}$		2.2×	2.6×	52.5	**73.1**	**52.8**	21.5	58.6	4.86
+ Conditional MoEs $_{\text{modality}}$		1.4×	1.0×	51.7	72.6	52.1	21.8	57.5	5.06
+ Conditional MoEs $_{\text{task}}$		1.4×	1.0×	**52.9**	**73.2**	**52.7**	21.2	**59.9**	**4.56**
+ Conditional MoEs $_{\text{attribute}}$		1.4×	1.0×	**52.8**	**73.3**	**53.1**	**23.0**	**60.0**	**4.56**

4.1 数据集

我们使用 Uni-Perceiver [93] 中相同的数据集来预训练我们的模型。具体而言，ImageNet-21k [18] 用于图像分类预训练。Kinetics-700 [37] 和 Moments in Time [57] 用于视频分类预训练。语言建模任务在 BookCorpus [94] 和英文维基百科（Books&Wiki）上训练。对于带有图像线索的语言建模和图像-文本检索，我们使用多个图像-文本配对数据集的组合：SBU Captions（SBU）[58]、Visual Genome [41]、COCO Caption [12]、CC3M [66]、CC12M [9] 和 YFCC [35]。遵循 Uni-Perceiver，我们利用 ImageNet1K [18]、Kinetics-400 [37]、COCO Caption [12] 和 Flickr30k [59] 来评估通用模型在下游任务上的性能。我们还使用两个用于评估对新任务泛化能力的数据集：MSVD [11] 和 GLUE [78]。此外，所有数据集许可证均包含在附录中。

4.2 实现细节

我们将普通的通用模型 Uni-Perceiver 与 Conditional MoEs 结合，用于三种不同变体的实验：Uni-Perceiver-Ti（Tiny）、Uni-Perceiver-B（Base）和 Uni-Perceiver-L（Large）。架构超参数请参见附录。若未特别说明，输入图像分辨率设置为 224×224。在每次训练迭代中，每个 GPU 独立采样一个任务和数据集。不同 GPU 的梯度在梯度反向传播后进行同步。我们使用 AdamW 优化器，基础学习率为 0.0005，权重衰减为 0.05。与 [52, 61] 类似，我们发现设置 $\beta_2 = 0.98$ 和 $\epsilon = 10^{-6}$ 有助于提高大规模训练时的稳定性。此外，使用 0.5 的梯度裁剪来稳定训练。

Uni-Perceiver-B 和 Uni-Perceiver-L 每隔一层配备 Conditional-MoEs 层，而 Uni-Perception-Ti 在所有层中使用 Conditional MoEs。按照 [64]，还在门控 logits 上使用了正态噪声，以便更好地探索新的潜在专家。如果没有特别说明，则使用 top-2 门控函数。关于 MoE 层的其他超参数，请参见附录。

4.3 消融实验

本部分探讨条件 MoE 是否能够有效缓解通用模型中的任务干扰，并比较不同的路由策略。表 3 总结了 Uni-Perceiver 及其变体在三个典型任务上的性能。与任务特定参数化相比，Uni-Perceiver 的性能下降证实了任务干扰的存在。采用任意路由策略引入条件 MoE 都可以缓解任务干扰问题，并显著提升性能。在这五种路由策略中，token 级、上下文级和模态级 MoE 的性能略差。我们认为，缺失的任务信息对于解决任务干扰至关重要。此外，数据依赖型 MoE，即 *i.e.,* token 级和上下文级 MoE，具有相对更高的训练和推理成本，而其他三种 MoE 通过使用重参数化技术具有出色的效率。尽管任务级 MoE 和属性 MoE 都取得了良好性能，但任务级 MoE 中专门的 task-id 设计使其难以泛化到新任务。因此，将使用采用属性路由策略的条件 MoE。

Table 4: The performance of incorporating Conditional MoEs with Uni-Perceiver on image classification, video classification and image-text retrieval. "#param" is the parameters required during model deployment. "#data" is the amount of visual training samples involved. "WT", "PT", and "FT" indicate w/o tuning, prompt tuning, and fine-tuning, respectively. "1%" and "100%" indicate the proportion of downstream data used for tuning. "$\text{FT}_{100\%}\uparrow$" means fine-tuning with larger image size. The subscript number next to score indicates that a different image resolution than 224 is used. † These methods use $> 20\times$ training data size and $> 10\times$ training cost than ours.

(a) Image Classification accuracy on ImageNet-1k.

Method	#param	#data	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	$\text{FT}_{100\%}\uparrow$
DeiT-B [76]	86M	1.28M	-	-	81.8	83.1_{384}
ViT-B [73]	86M	15.5M	-	-	84.0	85.5_{384}
ViT-L [73]	307M	15.5M	-	-	84.0	85.6_{384}
OFA [79]	472M	60.6M	-	-	-	84.9_{480}
CLIP [61]	307M	400M	76.2_{336}	-	-	-
†ALIGN [33]	480M	1.8B	76.4_{289}	-	-	88.6_{289}
†Florence [89]	637M	900M	83.7_{384}	-	-	$90.0_{\geq 384}$
†CoCa-B [87]	86M	4.8B	82.6_{576}	-	-	88.3_{576}
†CoCa-L [87]	303M	4.8B	84.8_{576}	-	-	90.2_{576}
†Flamingo-3B [3]	3.2B	2.3B	-	71.0_{320}	-	-
Uni-Perceiver-B	86M	44.1M	79.2	80.9	84.0	85.2_{384}
+ Conditional MoEs	86M	44.1M	80.3	82.0	84.5	85.8_{384}
Uni-Perceiver-L	303M	44.1M	82.7	84.2	86.2	87.0_{384}
+ Conditional MoEs	303M	44.1M	83.4	84.9	86.4	87.0_{384}

(b) Video classification accuracy on Kinetics-400.

Method	#param	#data	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$
TimeSformer-B [7]	121.4M	14.2M	-	-	80.7
VATT-B [2]	87.9M	238M	-	-	79.6_{320}
VATT-L [2]	306.1M	238M	-	-	82.1_{320}
ViViT-L [5]	>307M	14.2M	-	-	81.7
ViViT-L [5]	>307M	300M	-	-	84.9
†Florence [89]	647M	900M	-	-	86.5_{384}
†CoCa [87]	2.1B	4.8B	-	-	88.9_{576}
Uni-Perceiver-B	86M	44.1M	74.5	74.8	77.7
+ Conditional MoEs	86M	44.1M	76.8	77.2	79.3
Uni-Perceiver-L	303M	44.1M	79.5	80.0	81.9
+ Conditional MoEs	303M	44.1M	82.1	83.0	84.2

(c) Image-text retrieval R@1 performance.

Method			Flickr30K						MSCOCO Caption					
			Image → Text			Text → Image			Image → Text			Text → Image		
	#param	#data	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$
ImageBERT [60]	170M	10M	70.7	-	87.0	54.3	-	73.1	44.0	-	66.4	32.3	-	50.5
UNITER-B [13]	146M	9.6M	80.7	-	85.9	66.2		72.5	-	-	64.4	-	-	50.3
UNITER-L [13]	363M	9.6M	83.6	-	87.3	68.7	-	75.6	-	-	65.7	-	-	52.9
ViLT [39]	87M	9.7M	73.2	-	74.8	56.5	-	61.5	55.0	-	64.4	40.4	-	42.7
FLAVA [71]	215M	70M	67.7	-	-	65.2	-	-	42.7	-	-	38.4	-	-
CLIP [61]	417M	400M	88.0_{336}	-	-	68.7_{336}	-	-	58.4_{336}	-	-	37.8_{336}	-	-
†ALIGN	820M	1.8B	88.6_{289}	-	95.3_{289}	75.7_{289}	-	84.9_{289}	58.6_{289}	-	77.0_{289}	45.6_{289}	-	59.9_{289}
†Florence [89]	893M	900M	90.9_{384}	-	97.2_{384}	76.7_{384}	-	87.9_{384}	64.7_{384}	-	-	47.2_{384}	-	-
†CoCa-B [87]	383M	4.8B	89.8_{576}	-	-	76.8_{576}	-	-	63.8_{576}	-	-	47.5_{576}	-	-
†CoCa-L [87]	787M	4.8B	92.5_{576}	-	-	80.4_{576}	-	-	66.3_{576}	-	-	51.2_{576}	-	-
†Flamingo-3B [3]	3.2B	2.3B	89.3_{320}	-	-	79.5_{320}	-	-	65.9_{320}	-	-	48.0_{320}	-	-
Uni-Perceiver-B	124M	44.1M	82.3	91.0	92.7	71.1	76.0	77.5	64.9	68.4	69.8	50.7	51.9	53.9
+ Conditional MoEs	167M	44.1M	82.1	91.3	93.6	72.4	78.5	79.8	64.6	68.9	70.5	51.6	52.6	54.1
Uni-Perceiver-L	354M	44.1M	83.7	92.1	94.7	74.2	80.0	82.1	67.8	73.3	74.4	54.1	56.2	57.9
+ Conditional MoEs	505M	44.1M	83.6	92.4	94.1	75.9	80.6	83.7	67.9	73.3	74.7	55.3	57.1	58.3

4.4 Evaluation on Pre-training tasks

Large-scale training is conducted to verify the effectiveness of our method, we first evaluate it on tasks involved in pre-training. Specifically, we use widely-used Imagenet-1k [18] and Kinetics-400 [37] to evaluate image and video classification respectively, and use popular Flickr30k [59] and COCO Caption [12] to evaluate image caption and image-text retrieval.

Tab. 4 and Tab. 5a show the results on the four pre-training tasks. We see that Uni-perceiver with our Conditional MoEs consistently outperforms vanilla Uni-perceiver by a large margin. Without any tuning, our models achieve comparable performance with task-specific SOTAs trained with similar model size and training data size. Note that, our approach is a generalist model pretrained on a unified task formulation, while task-specific approaches are trained specifically for the target task.

When prompt tuned on only 1% downstream data, the performance of our models are boosted to a level close to counterparts that use $>50\times$ training data sizes and $>10\times$ training cost. For the prompt tuning of our models, only a small amount of parameters are tuned, and the encoder is still fixed and shared among different tasks, indicating that generalist models with Conditional MoEs can handle different tasks with significant low cost than counterparts.

We further fine-tune our models with 100% of the downstream data. In this case, our model achieves performance on-par with or better than the SOTAs trained with similar data size on all these tasks, which proves generalist models with Conditional MoEs has learned high-quality representations.

表4：在图像分类、视频分类和图文检索任务中，将条件 MoE 与 Uni-Perceiver 结合的性能。“#param”表示模型部署期间所需的参数量。“#data”表示涉及的视觉训练样本数量。“WT”、“PT”和“FT”分别表示不调优、提示调优和微调。“1%”和“100%”表示用于调优的下游数据比例。“$\text{FT}_{100\%}\uparrow$”表示使用更大图像尺寸进行微调。分数旁的下标数字表示使用了不同于 224 的图像分辨率。† 这些方法使用了比我们多 $>20\times$ 的训练数据规模和多 $>10\times$ 的训练成本。

(a) 在 ImageNet-1k 上的图像分类准确率。

Method	#param	#data	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	$\text{FT}_{100\%}\uparrow$
DeiT-B [76]	86M	1.28M	-	-	81.8	83.1_{384}
ViT-B [73]	86M	15.5M	-	-	84.0	85.5_{384}
ViT-L [73]	307M	15.5M	-	-	84.0	85.6_{384}
OFA [79]	472M	60.6M	-	-	-	84.9_{480}
CLIP [61]	307M	400M	76.2_{336}	-	-	-
†ALIGN [33]	480M	1.8B	76.4_{289}	-	-	88.6_{289}
†Florence [89]	637M	900M	83.7_{384}	-	-	$90.0_{\geq 384}$
†CoCa-B [87]	86M	4.8B	82.6_{576}	-	-	88.3_{576}
†CoCa-L [87]	303M	4.8B	84.8_{576}	-	-	90.2_{576}
†Flamingo-3B [3]	3.2B	2.3B	-	71.0_{320}	-	-
Uni-Perceiver-B	86M	44.1M	79.2	80.9	84.0	85.2_{384}
+ Conditional MoEs	86M	44.1M	80.3	82.0	84.5	85.8_{384}
Uni-Perceiver-L	303M	44.1M	82.7	84.2	86.2	87.0_{384}
+ Conditional MoEs	303M	44.1M	83.4	84.9	86.4	87.0_{384}

(b) Kinetics-400 上的视频分类准确率。

Method	#param	#data	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$
TimeSformer-B [7]	121.4M	14.2M	-	-	80.7
VATT-B [2]	87.9M	238M	-	-	79.6_{320}
VATT-L [2]	306.1M	238M	-	-	82.1_{320}
ViViT-L [5]	>307M	14.2M	-	-	81.7
ViViT-L [5]	>307M	300M	-	-	84.9
†Florence [89]	647M	900M	-	-	86.5_{384}
†CoCa [87]	2.1B	4.8B	-	-	88.9_{576}
Uni-Perceiver-B	86M	44.1M	74.5	74.8	77.7
+ Conditional MoEs	86M	44.1M	76.8	77.2	79.3
Uni-Perceiver-L	303M	44.1M	79.5	80.0	81.9
+ Conditional MoEs	303M	44.1M	82.1	83.0	84.2

(c) 图文检索 R@1 性能。

Method			Flickr30K						MSCOCO Caption					
			Image → Text			Text → Image			Image → Text			Text → Image		
	#param	#data	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$	WT	$\text{PT}_{1\%}$	$\text{FT}_{100\%}$
ImageBERT [60]	170M	10M	70.7	-	87.0	54.3	-	73.1	44.0	-	66.4	32.3	-	50.5
UNITER-B [13]	146M	9.6M	80.7	-	85.9	66.2		72.5	-	-	64.4	-	-	50.3
UNITER-L [13]	363M	9.6M	83.6	-	87.3	68.7	-	75.6	-	-	65.7	-	-	52.9
ViLT [39]	87M	9.7M	73.2	-	74.8	56.5	-	61.5	55.0	-	64.4	40.4	-	42.7
FLAVA [71]	215M	70M	67.7	-	-	65.2	-	-	42.7	-	-	38.4	-	-
CLIP [61]	417M	400M	88.0_{336}	-	-	68.7_{336}	-	-	58.4_{336}	-	-	37.8_{336}	-	-
†ALIGN	820M	1.8B	88.6_{289}	-	95.3_{289}	75.7_{289}	-	84.9_{289}	58.6_{289}	-	77.0_{289}	45.6_{289}	-	59.9_{289}
†Florence [89]	893M	900M	90.9_{384}	-	97.2_{384}	76.7_{384}	-	87.9_{384}	64.7_{384}	-	-	47.2_{384}	-	-
†CoCa-B [87]	383M	4.8B	89.8_{576}	-	-	76.8_{576}	-	-	63.8_{576}	-	-	47.5_{576}	-	-
†CoCa-L [87]	787M	4.8B	92.5_{576}	-	-	80.4_{576}	-	-	66.3_{576}	-	-	51.2_{576}	-	-
†Flamingo-3B [3]	3.2B	2.3B	89.3_{320}	-	-	79.5_{320}	-	-	65.9_{320}	-	-	48.0_{320}	-	-
Uni-Perceiver-B	124M	44.1M	82.3	91.0	92.7	71.1	76.0	77.5	64.9	68.4	69.8	50.7	51.9	53.9
+ Conditional MoEs	167M	44.1M	82.1	91.3	93.6	72.4	78.5	79.8	64.6	68.9	70.5	51.6	52.6	54.1
Uni-Perceiver-L	354M	44.1M	83.7	92.1	94.7	74.2	80.0	82.1	67.8	73.3	74.4	54.1	56.2	57.9
+ Conditional MoEs	505M	44.1M	83.6	92.4	94.1	75.9	80.6	83.7	67.9	73.3	74.7	55.3	57.1	58.3

4.4 预训练任务评估

为了验证我们方法的有效性，我们进行了大规模训练，并首先在预训练所涉及的任务上对其进行评估。具体而言，我们使用广泛采用的 ImageNet-1k [18] 和 Kinetics-400 [37] 分别评估图像分类和视频分类，并使用常用的 Flickr30k [59] 和 COCO Caption [12] 评估图像描述生成和图文检索。

表4和表5a展示了四个预训练任务的结果。可以看到，采用我们的条件 MoE 的 Uni-perceiver 始终以较大优势优于原始 Uni-perceiver。在不进行任何调优的情况下，我们的模型在性能上与使用相似模型规模和训练数据规模训练的任务特定 SOTA 方法相当。需要注意的是，我们的方法是在统一任务形式上预训练的通用模型，而任务特定方法则是专门针对目标任务进行训练的。

当仅在 1% 的下游数据上进行提示调优时，我们模型的性能提升到了接近使用 $>50\times$ 训练数据规模和 $>10\times$ 训练成本的对应模型的水平。对于我们模型的提示调优，仅需调优少量参数，编码器仍然保持固定并在不同任务之间共享，这表明具有条件 MoE 的通用模型能够以比对应模型显著更低的成本处理不同任务。

我们进一步使用 100% 的下游数据对模型进行微调。在这种情况下，我们的模型在所有这些任务上的性能均达到或超过了使用相似数据规模训练的 SOTA，这证明了采用 Conditional MoEs 的通用模型已经学习到了高质量的表示。

Table 5: The performance of incorporating Conditional MoEs with Uni-Perceiver on image caption, natural language understanding, video-text retrieval and video caption, where the last three tasks are not involved in pre-training.

(a) Image caption BLEU@4 performance. * means methods use region features as network inputs. ‡ indicates that Cider optimization is used.

Method	#param	data	MSCOCO Caption WT	$PT_{1\%}$	$FT_{100\%}$	Flickr30k WT	$PT_{1\%}$	$FT_{100\%}$
*Unified VLP [92]	86M	3.0M	-	-	36.5	-	-	30.1
*OSCAR-B [46]	154M	6.5M	-	-	36.5	-	-	-
*OSCAR-L [46]	384M	6.5M	-	-	37.4	-	-	-
UNICORN [86]	198M	200k	-	-	35.8	-	-	-
BLIP-B [44]	252M	129M	-	-	39.7_{384}	-	-	-
BLIP-L [44]	473M	129M	-	-	40.4_{384}	-	-	-
CLIP-VIL [69]	> 459M	400M	-	-	40.2	-	-	-
SimVLM [84]	632M	1.8B	11.2_{480}	-	40.6_{480}	-	-	-
*†OSCAR-L [46]	384M	6.5M	-	-	41.7	-	-	-
‡OFA [79]	472M	60.6M	-	-	43.5_{480}	-	-	-
†CoCa [87]	2.1B	4.8B			40.9_{576}	-	-	-
Uni-Perceiver-B	124M	44.1M	32.0	35.5	36.4	14.7	30.2	31.2
+ Conditional MoEs	167M	44.1M	33.2	36.8	37.3	15.9	30.7	32.4
Uni-Perceiver-L	354M	44.1M	35.3	38.6	39.2	15.1	32.9	35.5
+ Conditional MoEs	505M	44.1M	35.5	39.3	40.5	15.8	33.7	36.2

(b) Natural language understanding (novel task) fine-tuned on GLUE. $\text{BERT}_{\text{BASE}}$ records from [34]. VisualBERT and LXMERT record from [30]. *RoBERTa uses $10\times$ training text tokens than ours.

Method	MNLI (Acc)	QNLI (Acc)	QQP (F1)	RTE (Acc)	SST-2 (Acc)	MRPC (F1)	CoLA (Mcc)
LXMERT [75]	80.4	84.2	75.3	57.2	90.2	80.4	39.0
VisualBERT [45]	81.6	87.0	86.0	56.6	89.4	82.1	38.6
SimVLM-B [84]	83.4	88.6	87.2	63.9	90.9	84.4	46.7
BERT-B [78]	84.5	88.4	88.3	63.5	92.9	89.0	54.7
BERT-L [78]	86.6	92.3	91.3	70.4	93.2	88.0	60.6
OFA-B [79]	84.3	91.1	88.4	70.8	92.7	90.6	52.3
OFA-L [79]	86.6	92.8	88.9	73.6	94.7	91.4	53.1
*RoBERTa-B [52]	87.6	92.8	91.9	78.7	94.8	90.2	63.6
*RoBERTa-L [52]	90.2	94.7	92.2	86.6	96.4	90.9	68.0
Uni-Perceiver-B	79.7	87.3	86.7	71.1	89.3	86.0	43.1
+ Conditional MoEs	81.5	88.2	87.8	75.8	90.9	87.1	52.2
Uni-Perceiver-L	82.5	89.2	87.7	73.7	91.2	90.2	52.0
+ Conditional MoEs	85.7	91.9	89.5	78.4	93.4	91.2	57.4

(c) Video-text retrieval (novel task) Recall@1 and video caption (novel task) BLEU@4 performance on MSVD.

Method	#param	#data	Video → Text WT	$PT_{1\%}$	$FT_{100\%}$	Text → Video WT	$PT_{1\%}$	$FT_{100\%}$	Video Caption WT	$PT_{1\%}$	$FT_{100\%}$
CLIP2video [22]	132M	400M	-	-	58.7	-	-	47.0	-	-	-
HunYuan_tvr [56]	364M	400M	-	-	68.0	-	-	52.7	-	-	-
ORG-TRL [91]	86M	2.0M	-	-	-	-	-	-	-	-	54.3
Uni-Perceiver-B	124M	44.1M	50.3	62.7	62.8	38.7	43.8	45.8	22.6	59.5	63.3
+ Conditional MoEs	167M	44.1M	52.8	65.6	65.0	40.0	45.3	47.8	23.4	60.0	65.4
Uni-Perceiver-L	354M	44.1M	45.4	65.5	65.2	34.2	48.6	50.8	24.7	67.2	68.3
+ Conditional MoEs	505M	44.1M	45.7	66.4	67.6	41.9	50.3	52.3	24.6	67.6	68.9

4.5 Generalization to Novel Tasks

The generalization ability is the most attractive aspect of generalist models, while the dynamic sub-networks activation of Conditional MoEs should maintain this ability while mitigating task interference. To verify this, we conduct experiments on video caption, video-text retrieval, and natural language understanding tasks, which did not appear in pre-training. As shown in Tab. 5c, our Uni-Perceiver equipped with Conditional MoEs could generalize to video-related tasks very well. They can obtain reasonable zero-shot performance on those tasks and also perform better than vanilla Uni-Perceiver with a great margin. Moreover, Uni-Perceiver-MoEs can achieve comparable results to SOTA methods with similar training cost by further conducting prompt tuning with only 1% data. Beyond that, Conditional MoEs can significantly boost the performance of Uni-Perceiver on GLUE benchmarks (Tab. 5b), owing to its excellent ability to resolve task interference in generalist models.

5 Conclusion

In this paper, we propose Conditional MoEs to address the task-interference issue in generalist models. By sparsely activate sub-networks without introducing any task-specific designs, generalist models can be pre-trained on multiple tasks jointly without performance degradation, while keeping the generalization ablity to novel tasks. We incorporate Conditional MoEs with the recently proposed generalist model Uni-Perceiver. With prompt tuning on 1% downstream data, the proposed sparse generalist model achieves competitive performance with previous SOTAs using only $<5\%$ training data and $<10\%$ training cost. We hope this work can motivate further research in generalist models.

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表5：在图像描述、自然语言理解、视频-文本检索和视频描述任务中，将条件 MoE 与 Uni-Perceiver 结合的性能表现，其中后三个任务未参与预训练。

(a) 图像描述 BLEU@4 性能。* 表示方法使用区域特征作为网络输入。‡ 表示使用了 Cider 优化。

Method	#param	data	MSCOCO Caption			Flickr30k		
			WT	$PT_{1\%}$	$FT_{100\%}$	WT	$PT_{1\%}$	$FT_{100\%}$
*Unified VLP [92]	86M	3.0M	-	-	36.5	-	-	30.1
*OSCAR-B [46]	154M	6.5M	-	-	36.5	-	-	-
*OSCAR-L [46]	384M	6.5M	-	-	37.4	-	-	-
UNICORN [86]	198M	200k	-	-	35.8	-	-	-
BLIP-B [44]	252M	129M	-	-	39.7_{384}	-	-	-
BLIP-L [44]	473M	129M	-	-	40.4_{384}	-	-	-
CLIP-VIL [69]	> 459M	400M	-	-	40.2	-	-	-
SimVLM [84]	632M	1.8B	11.2_{480}	-	40.6_{480}	-	-	-
*‡OSCAR-L [46]	384M	6.5M	-	-	41.7	-	-	-
‡OFA [79]	472M	60.6M	-	-	43.5_{480}	-	-	-
†CoCa [87]	2.1B	4.8B			40.9_{576}	-	-	-
Uni-Perceiver-B	124M	44.1M	32.0	35.5	36.4	14.7	30.2	31.2
+ Conditional MoEs	167M	44.1M	33.2	36.8	37.3	15.9	30.7	32.4
Uni-Perceiver-L	354M	44.1M	35.3	38.6	39.2	15.1	32.9	35.5
+ Conditional MoEs	505M	44.1M	35.5	39.3	40.5	15.8	33.7	36.2

(b) 在 GLUE 上微调的自然语言理解（新任务）。$BERT_{BASE}$ 记录来自 [34]。VisualBERT 和 LXMERT 记录来自 [30]。*RoBERTa 使用的训练文本标记比我们多 10×。

Method	MNLI (Acc)	QNLI (Acc)	QQP (F1)	RTE (Acc)	SST-2 (Acc)	MRPC (F1)	CoLA (Mcc)
LXMERT [75]	80.4	84.2	75.3	57.2	90.2	80.4	39.0
VisualBERT [45]	81.6	87.0	86.0	56.6	89.4	82.1	38.6
SimVLM-B [84]	83.4	88.6	87.2	63.9	90.9	84.4	46.7
BERT-B [78]	84.5	88.4	88.3	63.5	92.9	89.0	54.7
BERT-L [78]	86.6	92.3	91.3	70.4	93.2	88.0	60.6
OFA-B [79]	84.3	91.1	88.4	70.8	92.7	90.6	52.3
OFA-L [79]	86.6	92.8	88.9	73.6	94.7	91.4	53.1
*RoBERTa-B [52]	87.6	92.8	91.9	78.7	94.8	90.2	63.6
*RoBERTa-L [52]	90.2	94.7	92.2	86.6	96.4	90.9	68.0
Uni-Perceiver-B	79.7	87.3	86.7	71.1	89.3	86.0	43.1
+ Conditional MoEs	81.5	88.2	87.8	75.8	90.9	87.1	52.2
Uni-Perceiver-L	82.5	89.2	87.7	73.7	91.2	90.2	52.0
+ Conditional MoEs	85.7	91.9	89.5	78.4	93.4	91.2	57.4

(c) MSVD 上视频-文本检索（新任务）的 Recall@1 和视频描述生成（新任务）的 BLEU@4 性能。

Method	#param	#data	Video → Text			Text → Video			Video Caption		
			WT	$PT_{1\%}$	$FT_{100\%}$	WT	$PT_{1\%}$	$FT_{100\%}$	WT	$PT_{1\%}$	$FT_{100\%}$
CLIP2video [22]	132M	400M	-	-	58.7	-	-	47.0	-	-	-
HunYuan_tvr [56]	364M	400M	-	-	68.0	-	-	52.7	-	-	-
ORG-TRL [91]	86M	2.0M	-	-	-	-	-	-	-	-	54.3
Uni-Perceiver-B	124M	44.1M	50.3	62.7	62.8	38.7	43.8	45.8	22.6	59.5	63.3
+ Conditional MoEs	167M	44.1M	52.8	65.6	65.0	40.0	45.3	47.8	23.4	60.0	65.4
Uni-Perceiver-L	354M	44.1M	45.4	65.5	65.2	34.2	48.6	50.8	24.7	67.2	68.3
+ Conditional MoEs	505M	44.1M	45.7	66.4	67.6	41.9	50.3	52.3	24.6	67.6	68.9

4.5 泛化到新任务

泛化能力是通用模型最具吸引力的方面，而条件 MoE 的动态子网络激活应在缓解任务干扰的同时保持这种能力。为验证这一点，我们在视频字幕、视频-文本检索和自然语言理解任务上进行了实验，这些任务并未出现在预训练中。如表 5c 所示，我们配备条件 MoE 的 Uni-Perceiver 能够很好地泛化到视频相关任务。它们可以在这些任务上获得合理的零样本性能，并且相比普通 Uni-Perceiver 也有大幅提升。此外，Uni-Perceiver-MoEs 仅使用 1% 的数据进一步进行提示调优，即可在相近训练成本下取得与 SOTA 方法相当的结果。除此之外，条件 MoE 还可以显著提升 Uni-Perceiver 在 GLUE 基准上的性能（表 5b），这得益于其在通用模型中解决任务干扰的出色能力。

5 结论

在本文中，我们提出了条件 MoE，以解决通用模型中的任务干扰问题。通过稀疏地激活子网络且不引入任何任务特定设计，通用模型可以在多个任务上联合预训练而不会出现性能下降，同时保持对新任务的泛化能力。我们将条件 MoE 与近期提出的通用模型 Uni-Perceiver 相结合。通过在 1% 下游数据上进行提示调优，所提出的稀疏通用模型仅使用 <5% 的训练数据和 <10% 的训练成本，就取得了与以往 SOTA 相竞争的性能。我们希望这项工作能够激励对通用模型的进一步研究。

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Table 6: Uni-Perceiver model variants used in this paper. "#vocab params" and "#encoder params" represent the number of vocabulary parameters and encoder network parameters, respectively.

	base model	embedding dimension	#heads	#layers	#vocab params	#encoder params	#total params
Uni-Perceiver-Ti	DeiT-Ti [76]	192	3	12	9.5M	5.3M	14.8M
Uni-Perceiver-B	ViT-B [20]	768	12	12	38M	86M	124M
Uni-Perceiver-L	ViT-L [20]	1024	16	24	51M	303M	354M

Table 7: Tasks and datasets used for our pre-training.

task	dataset	batch size/GPU	sampling weight	loss weight
Image Classification	ImageNet-21k [18]	220	0.2486	1.0
Video Classification	Kinetics-700 [37]	6	0.01	0.1
	Moments in Time [57]	24	0.02	0.1
Masked Language Modeling	Books&Wiki [94]	256	0.275	0.5
Image Caption	YFCC [35]	100	0.0584	1.0
	CC12M [9]	100	0.05057	1.0
	CC3M [66]	100	0.026295	1.0
	Visual Genome [41]	100	0.01766	1.0
	COCO Caption [12]	100	0.01144	1.0
	SBU [58]	100	0.01383	1.0
Image-Text Retrieval	YFCC [35]	160	0.0584	0.5
	CC12M [9]	160	05057	0.5
	CC3M [66]	160	0.026295	0.5
	Visual Genome [41]	160	0.01766	0.5
	COCO Caption [12]	160	0.01144	0.5
	SBU [58]	160	0.01383	0.5

6 Appendix

6.1 Experimental Details

Pre-training Details. Uni-Perceivers with three variants are used in our works, which are summarized in Tab. 6. Uni-Perceiver-Ti adopts the same setting of DeiT-Ti [76] in ablation experiments. Uni-Perceiver-B and Uni-Perceiver-L have the same architectures as their corresponding ViT variants, respectively. We follow most of the settings in Uni-Perceiver [93]: cross-entropy loss with label smoothing of 0.1 is adopted for all tasks, and the negative samples for retrieval tasks are only from the local batch in the current GPU. We also apply the same data augmentation techniques as Uni-Perceiver [93] to image and video modalities to avoid overfitting. The Uni-Perceiver and Uni-Perceiver-MoE models are pre-trained on 32 NVIDIA-A100 GPUs (80GB memory) for 400k iterations.

There are some setting changes to improve the training stability of the original Uni-Perceiver. Following [102], a uniform drop rate for stochastic depth is used across all encoder layers and are adapted according to the model size. Additionally, LayerScale [101] is used to facilitate the convergence of Transformer training, and the same initialization of 10^{-3} is set to all models for simplicity. Besides, Tab. 7 lists the batch size, sampling weight, and loss weight for each task and dataset in the pre-training stage. The loss weights are adjusted to meet reasonable optimizations for all tasks by observing the early training losses through short-epoch experiments. Those sampling weights for each task and dataset are proportional to the square root of the dataset size, which is demonstrated to be an effective heuristic to ease data imbalance across different datasets [4]. Following [102], we first pre-train with the image resolution of 160×160 and the patch size of 16×16, and continue pre-training for another 10% of total iterations on a higher resolution of 224×224. Furthermore, the implementation of mixed precision training in [100] is also employed to train Uni-Perceiver with a larger batch size. Based on the above settings, we can train Uni-Perceiver more efficiently. As shown in Tab. 8, our re-implemented Uni-Perceiver also achieves better performance across various tasks.

The number of experts in each Conditional-MoE layer is set to 8 by default. For Conditional MoEs with data-dependent routing strategies, *i.e.,* token-level and context-level, we use a capacity factor of 1.0 for the training stage and 2.0 for the evaluation stage [23, 40]. Besides, The balance loss in [23] is also employed for data-dependent MoEs to accomplish the balanced load of expert utilization, which is added to the total loss with a multiplicative coefficient of 10^{-2}.

表6：本文中使用的 Uni-Perceiver 模型变体。“#vocab params”和“#encoder params”分别表示词表参数数量和编码器网络参数数量。

	base model	embedding dimension	#heads	#layers	#vocab params	#encoder params	#total params
Uni-Perceiver-Ti	DeiT-Ti [76]	192	3	12	9.5M	5.3M	14.8M
Uni-Perceiver-B	ViT-B [20]	768	12	12	38M	86M	124M
Uni-Perceiver-L	ViT-L [20]	1024	16	24	51M	303M	354M

表 7：我们预训练所用的任务和数据集。

task	dataset	batch size/GPU	sampling weight	loss weight
Image Classification	ImageNet-21k [18]	220	0.2486	1.0
Video Classification	Kinetics-700 [37]	6	0.01	0.1
	Moments in Time [57]	24	0.02	0.1
Masked Language Modeling	Books&Wiki [94]	256	0.275	0.5
Image Caption	YFCC [35]	100	0.0584	1.0
	CC12M [9]	100	0.05057	1.0
	CC3M [66]	100	0.026295	1.0
	Visual Genome [41]	100	0.01766	1.0
	COCO Caption [12]	100	0.01144	1.0
	SBU [58]	100	0.01383	1.0
Image-Text Retrieval	YFCC [35]	160	0.0584	0.5
	CC12M [9]	160	05057	0.5
	CC3M [66]	160	0.026295	0.5
	Visual Genome [41]	160	0.01766	0.5
	COCO Caption [12]	160	0.01144	0.5
	SBU [58]	160	0.01383	0.5

6 附录

6.1 实验细节

预训练细节。我们的工作中使用了三种变体的 Uni-Perceiver，如表 6 所总结。Uni-Perceiver-Ti 在消融实验中采用与 DeiT-Ti [76] 相同的设置。Uni-Perceiver-B 和 Uni-Perceiver-L 分别与其对应的 ViT 变体具有相同的架构。我们遵循 Uni-Perceiver [93] 中的大部分设置：所有任务均采用标签平滑为 0.1 的交叉熵损失，检索任务的负样本仅来自当前 GPU 上的本地批次。我们还将与 Uni-Perceiver [93] 相同的数据增强技术应用于图像和视频模态，以避免过拟合。Uni-Perceiver 和 Uni-Perceiver-MoE 模型在 32 块 NVIDIA-A100 GPU（80GB 显存）上进行了 400k 次迭代的预训练。

为了提升原始 Uni-Perceiver 的训练稳定性，我们进行了一些设置调整。按照 [102]，在所有编码器层中使用统一的随机深度丢弃率，并根据模型规模进行调整。此外，使用 LayerScale [101] 来促进 Transformer 训练的收敛，并且为简化起见，所有模型都设置相同的 10^{-3} 初始化。除此之外，表 7 列出了预训练阶段每个任务和数据集的批大小、采样权重和损失权重。通过短 epoch 实验观察早期训练损失，对损失权重进行调整，以满足所有任务的合理优化需求。每个任务和数据集的采样权重与数据集大小的平方根成正比，这已被证明是一种有效的启发式方法，可缓解不同数据集之间的数据不平衡 [4]。按照 [102]，我们首先使用 160×160 的图像分辨率和 16×16 的 patch 大小进行预训练，然后在更高的 224×224 分辨率上继续进行总迭代次数 10% 的预训练。此外，还采用了 [100] 中的混合精度训练实现，以便使用更大的批大小训练 Uni-Perceiver。基于上述设置，我们可以更高效地训练 Uni-Perceiver。如表 8 所示，我们重新实现的 Uni-Perceiver 在各种任务上也取得了更好的性能。

每个 Conditional-MoE 层中的专家数量默认设置为 8。对于采用数据依赖路由策略的 Conditional MoE，即 *i.e.,* token 级和上下文级，我们在训练阶段使用 1.0 的容量因子，在评估阶段使用 2.0 的容量因子 [23, 40]。此外，[23] 中的均衡损失也用于数据依赖 MoE，以实现专家利用率的负载均衡，并以 10^{-2} 的乘法系数加入总损失中。

Table 8: The comparison between the original Uni-Perceiver-B [93] and our re-implemented Uni-Perceiver-B*. Results are reported for image classification accuracy on ImageNet-1K, video classification accuracy on Kinetics-400, image-text retrieval R@1 on Flickr30K, and image caption BLEU@4 on COCO Caption. 'TR' and 'IR' represent text retrieval and image retrieval, respectively.

	ImageNet-1k			Kinetics-400			Flickr30K		COCO Caption	pre-training time
	WT	$PT_{1\%}$	$FT_{100\%}$	WT	$PT_{1\%}$	$FT_{100\%}$	$TR(FT_{100\%})$	$IR(FT_{100\%})$	$FT_{100\%}$	TPU-v3-core-days
Uni-Perceiver-B [93]	78.0	80.2	83.8	73.5	73.6	75.8	87.9	74.9	35.6	$\sim$ 3.4k
Uni-Perceiver-B*	79.2	80.9	84.0	74.5	74.8	77.7	92.7	77.5	36.4	$\sim$ 1.0k

Table 9: Hyper-parameters for tuning on the downstream tasks. "*param1/param2*" denotes the corresponding parameters used for fine-tuning and prompt tuning, respectively.

	ImageNet-1k	Kinetics-400	COCO Caption	Flickr30K Caption	COCO Retrieval	Flickr30K Retrieval
Gradient clip	1.0/1.0					
Stoch. Depth	0.2/0.2					
Weight decay rate	1×10^{-4}/0.0					
LR decay schedule	Cosine Schedule Decaying to Zero/Constant Learning Rate					
Train steps	20k/5k	20k/2k	10k/1k	4k/200	10k/500	5k/100
Train batch size	2048/1024	64/32	512/512	512/512	2048/2048	2048/2048
Warm-up steps	2k/500	2k/200	1k/100	400/20	1k/50	500/10
Learning rate	$2\times10^{-5}/1\times10^{-3}$	$5\times10^{-6}/5\times10^{-4}$	$2\times10^{-5}/1\times10^{-3}$	$2\times10^{-5}/1\times10^{-3}$	$5\times10^{-6}/1\times10^{-3}$	$5\times10^{-6}/1\times10^{-3}$

Fine-tuning & Prompt Tuning. In addition to evaluation without any parameter tuning, fine-tuning with 100% data and prompt tuning with 1% data are also conducted to evaluate the model performance. We mainly follow the practice in Uni-Perceiver. Specifically, for prompt tuning, following P-Tuning v2 [97], learnable prompt tokens with random initialization are added at each encoder layer, and class labels with linear heads are added for classification tasks. The <SPE> token and layer norm parameters are also tuned. All training receipts for fine-tuning and prompt tuning are listed in Tab. 9.

Removing Overlap. Following [93], we carefully remove those videos overlapping with the validation set of Kinetics-400 in the training set of Kinetics-700.

6.2 The Placement of Conditional MoEs

Previous methods usually only incorporate the MoE layers with every other dense feed-forward network (FFN) layer [23, 40]. Conversely, we prefer using Conditional-MoE layers in every layer in the transformer, which is beneficial for mitigating task interference thoroughly. Tab. 11 shows the results when applying Conditional-MoE layers at intervals or only on certain type of layers, *i.e.,* FFN layers or self-attention layers. We can observe that the more layers are replaced with attribute-level MoEs, the more the task interference will be mitigated and the higher performance of each task can be achieved. Besides, applying Conditional MoEs in both self-attention and FFN layers can better alleviate the task interference. Nevertheless, only equipping self-attention layers with Conditional MoEs has limited ability to resolve the task interference.

6.3 Visualization of Gating decisions for Attribute-level MoEs

We show the expert distributions of different layers of a trained Uni-Perceiver-MoE model in Fig. 3. Both Conditional MoEs in self-attention layers (Fig. 3a) and FFN layers (Fig. 3b) have learnt to activate experts sparsely according to the token attributes. There are some experts shared by tokens with the same modality. For example, "images" for image classification and image caption tasks are usually routed to the same experts. Additionally, there are also some experts shared by the inputs and targets from the same task, *e.g.,* the expert *b* in the 7*th* FFN layer. Interestingly, the attribute of causal attention mask is also utilized by Conditional MoEs, *e.g.,* the 2*nd* and the 9*th* self-attention layers, indicating the potential interference between auto-encoding and auto-regressive tasks.

6.4 Training Statistics of Relevant Methods

Tab. 10 lists the training statistics of some methods relevant to our work. As discussed in Sec. 2, these methods are divided into three categories: specialized models, integrated specialized models, and generalist models. For a fair comparison, all of the data used to train each method from scratch is recorded, including the data used in the off-the-shelf pre-trained models. We can see that the training data scale of Uni-Perceiver-MoE is relatively much smaller than most of the generalist models and comparable with some specialized models.

表 8：原始 Uni-Perceiver-B [93] 与我们重新实现的 Uni-Perceiver-B* 之间的比较。结果分别报告了 ImageNet-1K 上的图像分类准确率、Kinetics-400 上的视频分类准确率、Flickr30K 上的图文检索 R@1，以及 COCO Caption 上的图像描述 BLEU@4。’TR’ 和 ’IR’ 分别表示文本检索和图像检索。

	ImageNet-1k			Kinetics-400			Flickr30K		COCO Caption	pre-training time
	WT	$PT_{1\%}$	$FT_{100\%}$	WT	$PT_{1\%}$	$FT_{100\%}$	$TR(FT_{100\%})$	$IR(FT_{100\%})$	$FT_{100\%}$	TPU-v3-core-days
Uni-Perceiver-B [93]	78.0	80.2	83.8	73.5	73.6	75.8	87.9	74.9	35.6	$\sim$ 3.4k
Uni-Perceiver-B*	79.2	80.9	84.0	74.5	74.8	77.7	92.7	77.5	36.4	$\sim$ 1.0k

表9：下游任务调优的超参数。“*param1/param2*”分别表示用于微调和提示调优的相应参数。

	ImageNet-1k	Kinetics-400	COCO Caption	Flickr30K Caption	COCO Retrieval	Flickr30K Retrieval
Gradient clip	1.0/1.0					
Stoch. Depth	0.2/0.2					
Weight decay rate	1×10^{-4}/0.0					
LR decay schedule	Cosine Schedule Decaying to Zero/Constant Learning Rate					
Train steps	20k/5k	20k/2k	10k/1k	4k/200	10k/500	5k/100
Train batch size	2048/1024	64/32	512/512	512/512	2048/2048	2048/2048
Warm-up steps	2k/500	2k/200	1k/100	400/20	1k/50	500/10
Learning rate	$2\times10^{-5}/1\times10^{-3}$	$5\times10^{-6}/5\times10^{-4}$	$2\times10^{-5}/1\times10^{-3}$	$2\times10^{-5}/1\times10^{-3}$	$5\times10^{-6}/1\times10^{-3}$	$5\times10^{-6}/1\times10^{-3}$

微调与提示调优。除了在不进行任何参数调优的情况下进行评估外，还进行了使用 100% 数据的微调和使用 1% 数据的提示调优，以评估模型性能。我们主要遵循 Uni-Perceiver 中的实践。具体而言，对于提示调优，遵循 P-Tuning v2 [97]，在每个编码器层添加随机初始化的可学习提示 token，并为分类任务添加带有线性头的类别标签。<SPE> token 和层归一化参数也会被调优。微调和提示调优的所有训练配置均列于表 9。

去除重叠。按照 [93]，我们仔细移除了 Kinetics-700 训练集中与 Kinetics-400 验证集重叠的那些视频。

6.2 条件式 MoE 的放置

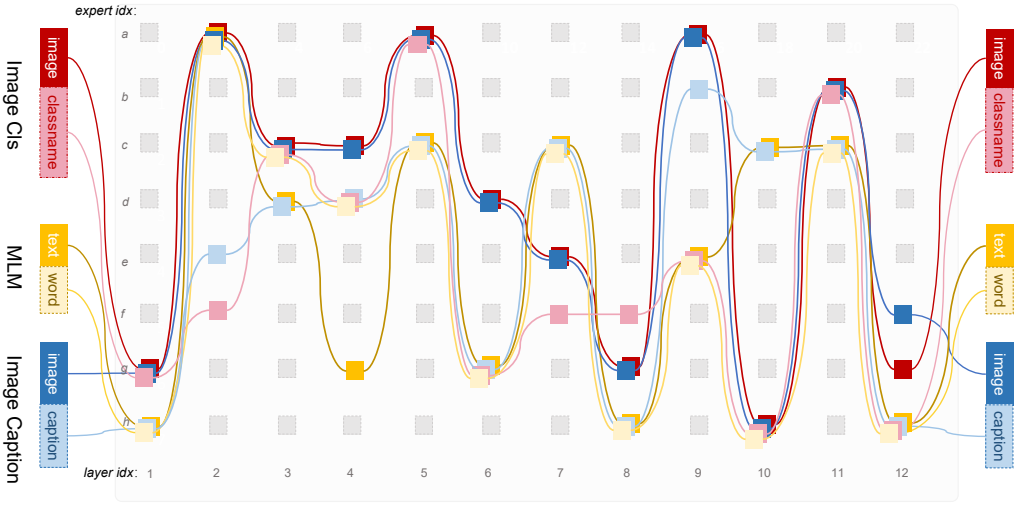
以往方法通常只在每隔一个密集前馈网络（FFN）层中引入 MoE 层 [23, 40]。相反，我们更倾向于在 Transformer 的每一层中使用 Conditional-MoE 层，这有助于更彻底地缓解任务干扰。表 11 展示了在间隔应用 Conditional-MoE 层，或仅在某一类型的层（*i.e.,* FFN 层或自注意力层）上应用时的结果。我们可以观察到，被属性级 MoE 替换的层越多，任务干扰就越能得到缓解，各项任务的性能也越高。此外，在自注意力层和 FFN 层中同时应用 Conditional MoE 能更好地缓解任务干扰。然而，仅为自注意力层配备 Conditional MoE 在解决任务干扰方面能力有限。

6.3 属性级 MoE 的门控决策可视化

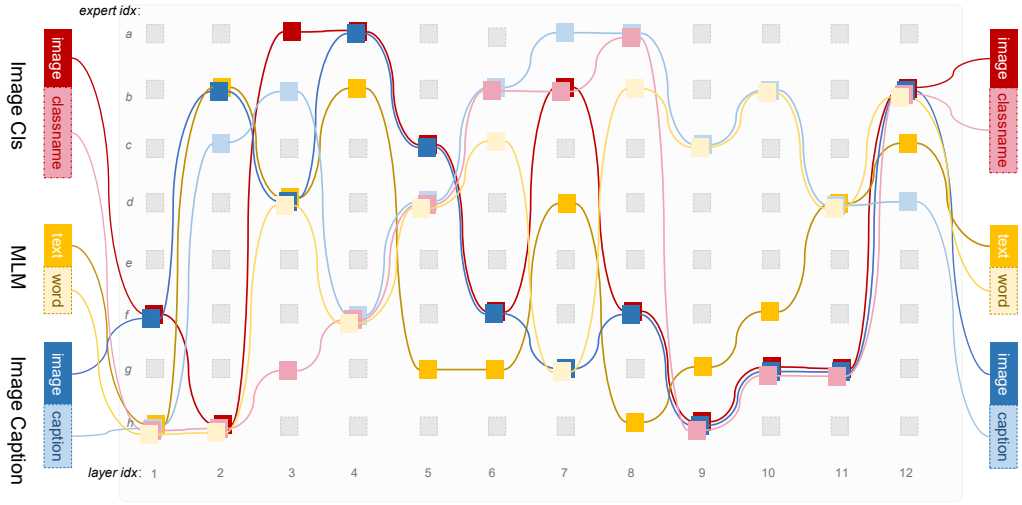
我们在图 3 中展示了训练后的 Uni-Perceiver-MoE 模型不同层的专家分布。自注意力层中的条件 MoE（图 3a）和 FFN 层中的条件 MoE（图 3b）都已学会根据词元属性稀疏地激活专家。一些专家会被具有相同模态的词元共享。例如，用于图像分类和图像描述任务的“图像”通常会被路由到相同的专家。此外，同一任务中的输入和目标也会共享一些专家，*e.g.,* 第 7*th* 个 FFN 层中的专家 *b*。有趣的是，因果注意力掩码这一属性也被条件 MoE 所利用，*e.g.,* 第 2*nd* 和第 9*th* 个自注意力层，这表明自编码任务和自回归任务之间可能存在潜在干扰。

6.4 相关方法的训练统计

表 10 列出了与我们工作相关的一些方法的训练统计数据。如第 2 节所述，这些方法分为三类：专用模型、集成式专用模型和通用模型。为了公平比较，我们记录了从零开始训练每种方法所使用的全部数据，包括现成预训练模型所使用的数据。可以看到，Uni-Perceiver-MoE 的训练数据规模相较于大多数通用模型要小得多，并且与一些专用模型相当。

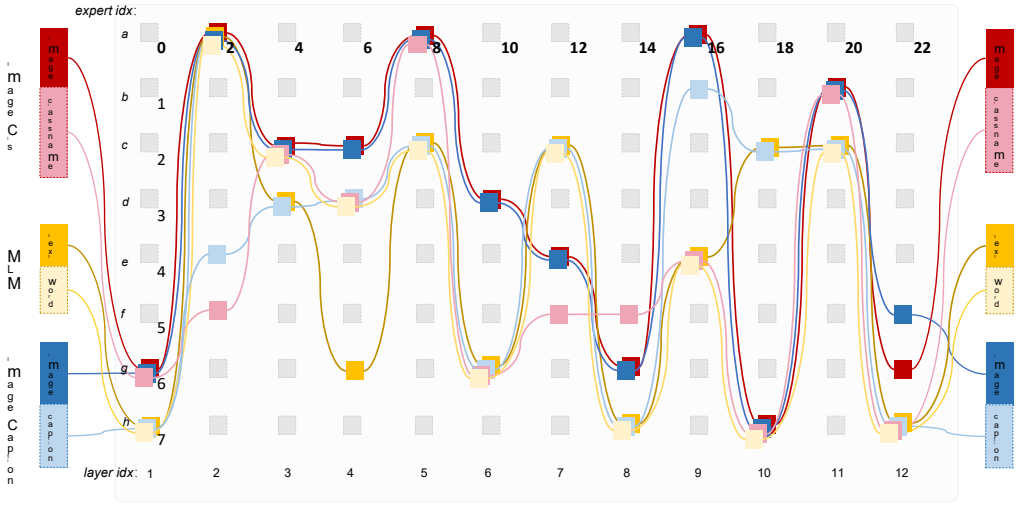


(a) Gating decisions of the self-attention layers for Uni-Perceiver-MoE-Ti.

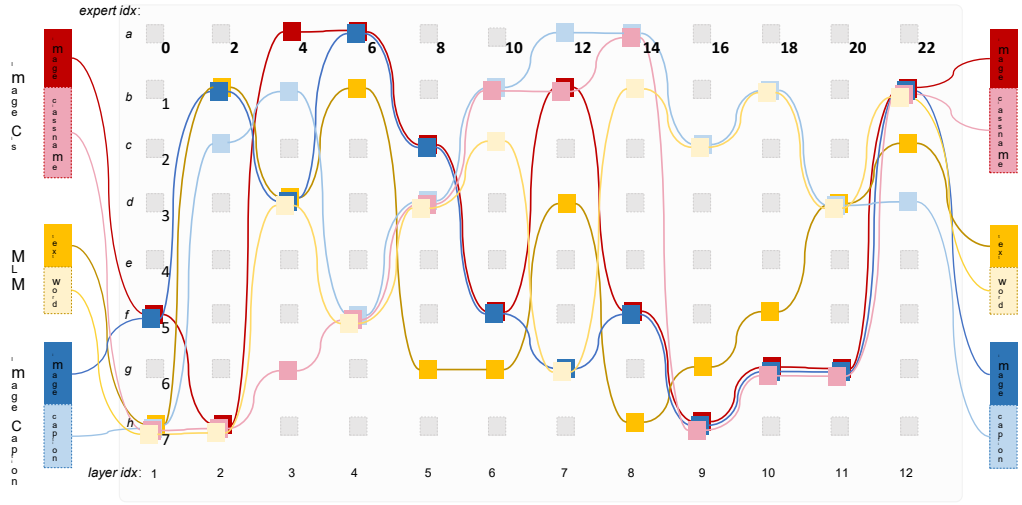


(b) Gating decisions of the FFN layers for Uni-Perceiver-MoE-Ti.

Figure 3: Gatinng decisions of the trained Uni-Perceiver-Ti with attribute-level MoEs. We show the top-1 activation of experts in every self-attention and FFN layer on three tasks: image classification (Image Cls) on ImageNet-1k, Masked Language Modeling (MLM) on Books&Wiki, and image caption on COCO Caption. The corresponding activation for the input and target from different tasks are highlighted with different colors. Grey shaded squares represent those experts that are not activated.



(a) Gating decisions of the self-attention layers for Uni-Perceiver-MoE-Ti.



(b) Uni-Perceiver-MoE-Ti 的 FFN 层的门控决策。

图 3：带有属性级 MoE 的训练后 Uni-Perceiver-Ti 的门控决策。我们展示了三个任务中每个自注意力层和 FFN 层的专家 top-1 激活：ImageNet-1k 上的图像分类（Image Cls）、Books &Wiki 上的掩码语言建模（MLM）以及 COCO Caption 上的图像描述。来自不同任务的输入和目标对应激活用不同颜色突出显示。灰色阴影方块表示未被激活的专家。

Table 10: Training statistics of relevant methods, which are categorized into specialized models, integrated specialized models, and generalist models. * represents the storage space of the plain texts. † represents the number of tokens in RL tasks. ¶ indicates the data used in the off-the-shelf pre-trained models. § datasets are private.

methods	training data			total visual data size	training time TPU-v3-core-days[1]
	dataset	data type	data size		
► *Specialized Models*					
BERT [19]	Books&Wiki	plain texts	16GB	-	1.4K
ORG-TRL [91]	ImageNet-1K	images	1.28M¶	1.6M	-
	Kinetics-400	videos	300K¶		
	MSCOCO	image-bounding boxes	118K¶		
	Books&Wiki	plain texts	16GB*¶		
Unified VLP [92]	CC3M	image-text pairs	3.0M	3.1M	-
	VG	image-bounding boxes	108K¶		
	Books&Wiki	plain texts	16GB*¶		
OSCAR [46]	COCO, CC3M, SBU, Flickr30K, VQA, GQA, VG-QA	image-text pair	6.5M	6.6M	-
	VG	image-bounding boxes+attributes	108K¶		
UNITER[13]	CC3M, SBU, COCO, VG	image-text pairs	9.6M	9.7M	0.5K
	VG	image-bounding boxes+attributes	108K¶		
ImageBERT [60]	CC3M, SBU, LAIT§	image-text pairs	13.7M	13.8M	-
	VG	image-bounding boxes	108K¶		
	Books&Wiki	plain texts	16GB¶		
TimeSformer [7]	ImageNet-21K	images	14.2M¶	14.5M	-
	Kinetics-400	videos	300K		
ViT-L [73]	ImageNet-21K, ImageNet-1K	images	15.5M	15.5M	0.23K
ViLT [39]	COCO, VG, CC3M, SBU	image-text pairs	9.7M	25.2M	-
	ImageNet-21K, ImageNet-1K	images	15.5M¶		
VATT [2]	AudioSet	audios	2.1M	138M	1.5K
	HowTo100M	video-audio-text triplets	136M		
BLIP [44]	COCO, VG, SBU CC3M, CC12M, LAION	image-text pairs	130M	144M	-
	ImageNet-21k	images	14.2M¶		
	Books&Wiki	plain texts	16GB*¶		
ViViT [5]	JFT-300M§, ImageNet-21K	images	300M¶	300.3M	-
	Kinetics-400	videos	300k		
CLIP [61]	CLIP data§	image-text pairs	400M	400M	18K
HunYuan_tvr [56]	CLIP data§	image-text pairs	400M	400M	-
CLIP2video [22]	CLIP data§	image-text pairs	400M	400M	-
CLIP-VIL [69]	CLIP data§	image-text pairs	400M	400M	-
ALIGN [33]	ALIGN data§	image-text pairs	1.8B	1.8B	-
► *Integrated Specialized Models*					
FLAVA [71]	COCO, SBU, Localized Narratives, CC3M, VG, Wikipedia Image Text, CC12M, Red Caps, YFCC	image-text pairs	70M	71.3M	-
	ImageNet-1K	images	1.28M		
	CCNews, BookCorpus	plain texts	970GB*		
Florence [89]	FLB-900M§	image-text pairs	900M	909M	44K
	FLOD-9M§	image-bounding boxes	9M		
► *Generalist Models*					
UNICORN [86]	COCO, VG, CC3M, SBU	image-text pairs	9.8M	25.3M	-
	ImageNet-21K, ImageNet-1K	images	15.5M¶		
	BooksCorpus, CC-News, OpenwebText, Stories	plain texts	160GB¶		
OFA [79]	CC12M, CC3M, SBU, COCO, VG-Cap	image-text pairs	15.25M	60.6M	-
	VQAv2, VG-QA, GQA	visual question answering	2.92M		
	RefCOCO, RefCOCO+, RefCOCOg, VG-Cap	image-instance-text triplets	3.2M		
	OpenImages, Object365, VG, COCO	image-bounding boxes	3.0M		
	YFCC100M, ImageNet-21K	Images	36.27M		
	Pile	plain texts	140GB*		
SimVLM [84]	ALIGN data§	image-text pairs	1.8B	1.8B	-
	Colossal Clean Crawled Corpus	plain texts	800GB*		
Gato [62]	DM Lab, ALE Atari, ALE Atari Extended, Sokoban, BabyAI, DM Control Suite, Procgen Benchmark, RGB Stacking simulator, RGB Stacking real robot, Meta-World, DM Manipulation Playground, Playroom	simulated data for RL tasks	1.5T†	2.16B	2K
	M3W§, ALIGN data§, CC3M, COCO LTIP§, QKVQA, VQAV2	image-text pairs	2.16B		
	MassiveWeb§	plain texts	1.9TB*		
Flamingo [3]	M3W§, ALIGN data§, LTIP§	image-text pairs	2.3B	2.3B	126K
	VTP§	video-text pair	27M		
CoCa [87]	JFT-3B§, ALIGN data§	image-text pairs	4.8B	4.8B	56K
Uni-Perceiver [93] & Uni-Perceiver-MoE	CC3M, CC12M, SBU, COCO, VG, YFCC	image-text pairs	28.6M	44.1M	4.2K
	ImageNet-21K	images	14.2M		
	Kinetics-700, Moments in Time	videos	1.33M		
	Books&Wiki	plain texts	16GB*		

[1] We convert the training time to the TPU-v3-core-days based on the TFLOPS of accelerators, *i.e.,* 1.0 TPU-v3-core-days ≈ 0.364 TPU-v4-core-days ≈ 0.151 NVIDIA-A100-days ≈ 0.302 NVIDIA-V100-days..

表 10：相关方法的训练统计信息，这些方法分为专用模型、集成式专用模型和通用模型。* 表示纯文本的存储空间。† 表示 RL 任务中的 token 数量。¶ 表示现成预训练模型中使用的数据。§ 数据集为私有数据集。

methods	training data			total visual data size	training time TPU-v3-core-days[1]
	dataset	data type	data size		
► *Specialized Models*					
BERT [19]	Books&Wiki	plain texts	16GB	-	1.4K
ORG-TRL [91]	ImageNet-1K	images	1.28M¶	1.6M	-
	Kinetics-400	videos	300K¶		
	MSCOCO	image-bounding boxes	118K¶		
	Books&Wiki	plain texts	16GB*¶		
Unified VLP [92]	CC3M	image-text pairs	3.0M	3.1M	-
	VG	image-bounding boxes	108K¶		
	Books&Wiki	plain texts	16GB*¶		
OSCAR [46]	COCO, CC3M, SBU, Flickr30K, VQA, GQA, VG-QA	image-text pair	6.5M	6.6M	-
	VG	image-bounding boxes+attributes	108K¶		
UNITER[13]	CC3M, SBU, COCO, VG	image-text pairs	9.6M	9.7M	0.5K
	VG	image-bounding boxes+attributes	108K¶		
ImageBERT [60]	CC3M, SBU, LAIT§	image-text pairs	13.7M	13.8M	-
	VG	image-bounding boxes	108K¶		
	Books&Wiki	plain texts	16GB¶		
TimeSformer [7]	ImageNet-21K	images	14.2M¶	14.5M	-
	Kinetics-400	videos	300K		
ViT-L [73]	ImageNet-21K, ImageNet-1K	images	15.5M	15.5M	0.23K
ViLT [39]	COCO, VG, CC3M, SBU	image-text pairs	9.7M	25.2M	-
	ImageNet-21K, ImageNet-1K	images	15.5M¶		
VATT [2]	AudioSet	audios	2.1M	138M	1.5K
	HowTo100M	video-audio-text triplets	136M		
BLIP [44]	COCO, VG, SBU CC3M, CC12M, LAION	image-text pairs	130M	144M	-
	ImageNet-21k	images	14.2M¶		
	Books&Wiki	plain texts	16GB*¶		
ViViT [5]	JFT-300M§, ImageNet-21K	images	300M¶	300.3M	-
	Kinetics-400	videos	300k		
CLIP [61]	CLIP data§	image-text pairs	400M	400M	18K
HunYuan_tvr [56]	CLIP data§	image-text pairs	400M	400M	-
CLIP2video [22]	CLIP data§	image-text pairs	400M	400M	-
CLIP-VIL [69]	CLIP data§	image-text pairs	400M	400M	-
ALIGN [33]	ALIGN data§	image-text pairs	1.8B	1.8B	-
► *Integrated Specialized Models*					
FLAVA [71]	COCO, SBU, Localized Narratives, CC3M, VG, Wikipedia Image Text, CC12M, Red Caps, YFCC	image-text pairs	70M	71.3M	-
	ImageNet-1K	images	1.28M		
	CCNews, BookCorpus	plain texts	970GB*		
Florence [89]	FLB-900M§	image-text pairs	900M	909M	44K
	FLOD-9M§	image-bounding boxes	9M		
► *Generalist Models*					
UNICORN [86]	COCO, VG, CC3M, SBU	image-text pairs	9.8M	25.3M	-
	ImageNet-21K, ImageNet-1K	images	15.5M¶		
	BooksCorpus, CC-News, OpenwebText, Stories	plain texts	160GB¶		
OFA [79]	CC12M, CC3M, SBU, COCO, VG-Cap	image-text pairs	15.25M	60.6M	-
	VQAv2, VG-QA, GQA	visual question answering	2.92M		
	RefCOCO, RefCOCO+, RefCOCOg, VG-Cap	image-instance-text triplets	3.2M		
	OpenImages, Object365, VG, COCO	image-bounding boxes	3.0M		
	YFCC100M, ImageNet-21K	Images	36.27M		
	Pile	plain texts	140GB*		
SimVLM [84]	ALIGN data§	image-text pairs	1.8B	1.8B	-
	Colossal Clean Crawled Corpus	plain texts	800GB*		
Gato [62]	DM Lab, ALE Atari, ALE Atari Extended, Sokoban, BabyAI, DM Control Suite, Procgen Benchmark, RGB Stacking simulator, RGB Stacking real robot, Meta-World, DM Manipulation Playground, Playroom	simulated data for RL tasks	1.5T†	2.16B	2K
	M3W§, ALIGN data§, CC3M, COCO LTIP§, QKVQA, VQAV2	image-text pairs	2.16B		
	MassiveWeb§	plain texts	1.9TB*		
Flamingo [3]	M3W§, ALIGN data§, LTIP§	image-text pairs	2.3B	2.3B	126K
	VTP§	video-text pair	27M		
CoCa [87]	JFT-3B§, ALIGN data§	image-text pairs	4.8B	4.8B	56K
Uni-Perceiver [93] & Uni-Perceiver-MoE	CC3M, CC12M, SBU, COCO, VG, YFCC	image-text pairs	28.6M	44.1M	4.2K
	ImageNet-21K	images	14.2M		
	Kinetics-700, Moments in Time	videos	1.33M		
	Books&Wiki	plain texts	16GB*		

1我们根据加速器的 TFLOPS 将训练时间转换为 TPU-v3-core-days，*i.e.,* 1.0 TPU-v3-core-days ≈ 0.364 TPU-v4-core-days ≈ 0.151 NVIDIA-A100-days ≈ 0.302 NVIDIA-V100-days..

Table 11: The performance of different settings of attribute-level routings. FFN MoE and SA MoE indicate whether Conditional MoEs are applied in FFN layers and self-attention layers, respectively. every-n means Conditional-MoE layers are placed every n layers in the Transformer encoder.

model	FFN MoE	SA MoE	every-n	ImageNet-1k		COCO Caption		MLM	
				$\uparrow \text{acc}_{\text{train}}$	$\uparrow \text{acc}_{\text{val}}$	$\uparrow \text{acc}_{\text{train}}$	$\uparrow \text{B@4}_{\text{val}}$	$\uparrow \text{acc}_{\text{train}}$	$\downarrow \text{ppl}_{\text{val}}$
Uni-Perceiver-Ti	-	-	-	47.3	68.3	49.2	18.2	54.5	5.86
+ Conditional MoEs $_{\text{attribute}}$	✓	✓	4	51.7	71.6	51.3	20.9	56.1	5.47
+ Conditional MoEs $_{\text{attribute}}$	✓	✓	2	52.2	72.3	51.8	21.1	57.3	5.13
+ Conditional MoEs $_{\text{attribute}}$	✓	✗	1	51.5	71.3	52.2	21.0	58.5	4.86
+ Conditional MoEs $_{\text{attribute}}$	✗	✓	1	49.2	69.7	51.0	20.6	55.8	5.50
+ Conditional MoEs $_{\text{attribute}}$	✓	✓	1	52.8	73.3	53.1	23.0	60.0	4.56

6.5 Licenses of Datasets

ImageNet-21K [18] is subject to the ImageNet terms of use [103].

Kinetics-700 & **Kinetics-400** [37] The kinetics dataset is licensed by Google Inc. under a Creative Commons Attribution 4.0 International License.

BooksCorpus [94] Replicate Toronto BookCorpus is open-source and licensed under GNU GPL, Version 3.

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COCO Caption [12] The images are subject to the Flickr terms of use [95].

SBU Caption [58] The images are subject to the Flickr terms of use [95].

Appendix References

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[96] R. Krishna. Visual genome terms & conditions of use. `https://visualgenome.org/about`.

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[99] G. LLC. Conceptual captions terms & conditions of use. `https://github.com/google-research-datasets/conceptual-captions/blob/master/LICENSE`, .

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[102] H. Touvron, M. Cord, and H. Jégou. Deit iii: Revenge of the vit. *arXiv preprint arXiv:2204.07118*, 2022.

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表 11：不同属性级路由设置的性能。FFN MoE 和 SA MoE 分别表示是否在 FFN 层和自注意力层中应用 Conditional MoE。every-*n* 表示在 Transformer 编码器中每隔 *n* 层放置一个 Conditional-MoE 层。

model	FFN MoE	SA MoE	every-*n*	ImageNet-1k		COCO Caption		MLM	
				$\uparrow \text{acc}_{\text{train}}$	$\uparrow \text{acc}_{\text{val}}$	$\uparrow \text{acc}_{\text{train}}$	$\uparrow \text{B@4}_{\text{val}}$	$\uparrow \text{acc}_{\text{train}}$	$\downarrow \text{ppl}_{\text{val}}$
Uni-Perceiver-Ti	-	-	-	47.3	68.3	49.2	18.2	54.5	5.86
+ Conditional MoEs $_{\text{attribute}}$	✓	✓	4	51.7	71.6	51.3	20.9	56.1	5.47
+ Conditional MoEs $_{\text{attribute}}$	✓	✓	2	52.2	72.3	51.8	21.1	57.3	5.13
+ Conditional MoEs $_{\text{attribute}}$	✓	✗	1	51.5	71.3	52.2	21.0	58.5	4.86
+ Conditional MoEs $_{\text{attribute}}$	✗	✓	1	49.2	69.7	51.0	20.6	55.8	5.50
+ Conditional MoEs $_{\text{attribute}}$	✓	✓	1	52.8	73.3	53.1	23.0	60.0	4.56

6.5 数据集许可证

ImageNet-21K [18] 受 ImageNet 使用条款 [103] 约束。

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